Internal and External R&D: An analysis of costs and benefits [*]

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Abstract

This paper analyzes the costs and benefits of in-house and contracted-out R&D. Estimating a dynamic model on Dutch ICT firms from 2000 to 2020, I find that productivity gains are similar across R&D activities, but that contracting R&D out is much more costly than doing it in-house. This explains why so few firms only contract out. However, for a firm that also does in-house R&D, 94% of this extra cost disappears, which I interpret as a hold-up problem in the external contract. Finally, a uniform subsidy raises R&D participation and firm value more than a subsidy for in-house R&D only.

Keywords: Research and Development, Complementarity, Innovation, Productivity, Internal R&D, External R&D, Outsourcing

JEL Codes: O31, O32, L25, C61, H25

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1 Introduction

Firms typically conduct R&D either in-house or by contracting it out to strategic partners within or outside their home country. Many firms also combine the two. In the Dutch ICT industry, about a third of firms perform only in-house R&D, and more than a quarter perform in-house R&D while contracting part of their research out, but only about 3% rely on contracted-out R&D alone. At the same time, R&D policy often treats the two modes differently. For example, the Dutch Innovation Box gives a lower tax rate to the profit from R&D, but under the OECD Nexus Approach, only a limited part of outsourced R&D still qualifies. This paper asks why so few firms rely on contracted-out R&D alone, why so many firms combine the two modes, and what this means for R&D subsidies. To answer these questions, I build and estimate a dynamic model of the firm's choice between in-house and contracted-out R&D, using Dutch microdata on ICT firms from 2000 to 2020.

I first document several stylized facts. First, the share of firms engaging in at least one type of R&D activity has increased over time, especially after the expansion of the Innovation Box in 2010. Most of these firms conduct R&D in-house, either exclusively or in combination with external partners. Second, the share of firms performing only in-house R&D has risen steadily, while the share relying only on contracted-out R&D has remained small and roughly constant. Third, R&D choices are persistent: a firm's R&D mode in one period strongly predicts its mode in the next. I also report a reduced-form test of whether combining the two modes raises value added or the share of new products by more than adding them separately. The test is inconclusive, and since it only compares outcomes across modes, it cannot tell whether firms combine the two modes because of their benefits or because of their costs.

To rationalize these facts, I construct a model that captures the costs and benefits of internal R&D (conducted in-house) and external R&D (contracted out to other firms, universities, or partners). The model has two parts. In the static part, firms compete under monopolistic competition, and a more productive firm earns a higher profit. In the dynamic part, in each period, a firm chooses one of four R&D modes: no R&D, in-house R&D only, contracted-out R&D only, or both. Each mode has its own cost, and the chosen mode shifts the firm's future productivity, and through it, its future profit. The model also allows doing both to be cheaper, or more expensive, than the sum of the two costs. I call this the cost-complementarity of in-house and contracted-out R&D. To recover these costs, I first estimate the elasticity of demand and the firm's productivity from the static part of the model. I then estimate the cost of each R&D mode from the firms' observed choices, using a dynamic discrete-choice estimator in the spirit of Rust (1987).

Three patterns emerge from this exercise. First, the productivity gains from in-house R&D, contracted-out R&D, and both are similar, about 5 to 6% relative to doing no R&D. Second, contracted-out R&D is much more costly than in-house R&D. In the full sample, contracted-out R&D costs about 2.45 million EUR, against 0.13 million EUR for in-house R&D. Before the Innovation Box expan-

sion, contracted-out R&D is about four times as costly as in-house R&D. This high cost is among the main reasons why so few firms rely on contracted-out R&D alone. Third, doing both is much cheaper than the sum of the two costs. For a firm that already does in-house R&D, about 94% of the extra cost of contracting out disappears, so doing both costs only about 0.16 million EUR more than doing in-house R&D alone.

I interpret these costs through a hold-up problem on the external R&D contract, in the spirit of Grossman and Hart (1986) and Aghion and Tirole (1994). When a firm contracts research out, it has to invest in the collaboration, and once it has done so, the partner can capture part of the value of the results. A firm without its own R&D staff therefore keeps only part of that value and invests too little, which makes contracting out costly. A firm with its own R&D staff is in a stronger position, since it can specify and evaluate the partner's work, and it keeps most of the value. Since the productivity gains are similar across modes, the complementarity between the two modes has to come from the cost side, which is where the hold-up problem operates.

Finally, I use the estimated model to simulate several R&D subsidies, which approximate the Dutch scheme as reductions in R&D costs. A uniform 5% cut in all R&D costs lowers the share of firms doing no R&D by 1.6 percentage points and raises firm value by about 2.7%. A 20% cut in the cost of in-house R&D only, in the spirit of the Dutch scheme's tilt toward in-house R&D, lowers that share by only 0.6 percentage points and raises firm value by about 1.0%. A 5% cut in the cost of contracted-out R&D only has effects close to the uniform cut. When cost-complementarity is shut off, the share of firms doing both falls sharply, and the gains in firm value from all three subsidies shrink to about 1% or less.

Related Literature.—This paper contributes to several strands of literature. First, it relates to the empirical literature on whether internal and external R&D are complementary. Cassiman and Veugelers (2006) test complementarity between internal and external R&D using insights from Arora (1996) and Athey and Stern (1998). Veugelers (1997) and Veugelers and Cassiman (1999) show that external technology sourcing raises in-house R&D when the firm already has its own R&D department, and that larger firms combine the two. Grimpe and Kaiser (2010) document an inverted-U relationship between R&D outsourcing and innovation, which is moderated by internal R&D. Related work includes Mohnen and Roeller (2005), who consider several innovation activities, Ceccagnoli and Jiang (2013) and Ceccagnoli et al. (2014), who study the firm-level drivers of complementarity between internal R&D and external knowledge acquisition, Berchicci (2013), who examines the innovative performance of R&D outsourcing, and Amoroso (2017), who analyzes complementarities among internal, external, and cooperative R&D. My reduced-form evidence uses a panel random-effects Tobit model, similar to Hagedoorn and Wang (2012) and Love et al. (2014), and exploits the panel structure of the data, whereas Cassiman and Veugelers (2006) use a single wave of the CIS. I find that the reduced form alone is inconclusive, so I estimate the complementarity on the cost side within a model.

Second, the paper relates to models of R&D and productivity. The static part of my model follows

Aw et al. (2011), Boler et al. (2015), and Peters et al. (2017). For productivity estimation, much of the literature relies on value-added output (e.g., Olley and Pakes, 1996; Levinsohn and Petrin, 2003; De Loecker and Warzynski, 2012; Ackerberg et al., 2015), and these approaches can face identification issues with gross output, which Gandhi et al. (2020) address with a nonparametric approach. I follow the semiparametric approach of Aw et al. (2011), Doraszelski and Jaumandreu (2013), and Peters et al. (2017). The dynamic part of my model is a single-agent dynamic discrete choice model in the tradition of Rust (1987), and I estimate it with his nested fixed-point algorithm. The idea of describing innovation through the cost of each action follows Igami (2017) and Igami and Uetake (2020), who estimate the costs of innovation from firms' innovation decisions in dynamic oligopoly models. Unlike their models, I do not model strategic interactions between firms, since I do not observe prices, and I explicitly incorporate complementarity between the two R&D modes. For this, I borrow insights from Miravete and Pernias (2006), who study complementarity between product and process innovation, and apply them to the cost side of my dynamic model. Akcigit and Kerr (2018) develop an endogenous growth model in which firms innovate internally, by improving their own product lines, and externally, by acquiring new product lines, where external innovation is more costly and scales less with firm size. In this paper, external R&D instead means contracting research out.

Third, the paper relates to the theory of firm boundaries in R&D. Goyal et al. (2008) derive conditions under which the projects of a firm are complementary, and show that complementarity can arise from cost advantages alone, even without spillovers from outside the firm. I find that the complementarity between in-house and contracted-out R&D comes from the cost side, since the productivity gains are similar across modes. I interpret the costs through a hold-up problem on the external contract (Grossman and Hart, 1986; Hart and Moore, 1990), applied to research contracts by Aghion and Tirole (1994), in which in-house R&D strengthens the firm's position in the contract. This interpretation is related to the idea of absorptive capacity (Cohen and Levinthal, 1989, 1990), in which in-house R&D helps the firm make use of external knowledge. Haggling over incomplete contracts (Bajari and Tadelis, 2001; Tadelis, 2002) produces the same pattern of costs. More broadly, I draw on the transaction-cost view of the boundary of the firm (Coase, 1937; Williamson, 1975). Jungbauer et al. (2026) study product-level outsourcing when an innovator risks cannibalizing its existing products, and Howells (2008) discusses the blurring boundaries of the firm in global R&D outsourcing.

Finally, the paper relates to the literature on R&D subsidies. Galaasen and Irarrazabal (2021) show that a uniform subsidy to Norwegian firms boosts investment, growth, and welfare, while a size-dependent subsidy raises aggregate R&D but reduces growth and welfare. Similarly, I find that a uniform subsidy raises the share of R&D-active firms by more than a subsidy that only lowers the cost of in-house R&D. The literature on fiscal incentives finds that R&D tax credits lower the user research cost (Hall and Van Reenen, 2000; Bloom et al., 2002). For the Netherlands, Lokshin and Mohnen (2012) find that the WBSO raises firms' R&D capital. Gonzalez et al. (2005) find that subsidies induce Spanish firms that do not perform R&D to start investing in it, and Bilir and

Morales (2020) find that, for U.S. multinationals, parent R&D spills over to foreign affiliates, so that about 20% of the returns to domestic R&D subsidies are realized abroad.

2 Institutional Setting, Data, and Motivating Evidence

This section outlines the Dutch Tax Incentives for Innovation, describes the data used, and presents stylized facts as motivating evidence.

2.1 Dutch Tax Incentives for Innovation

To promote research and development in the Netherlands, the Dutch government introduced a comprehensive innovation program that provides tax incentives for Dutch companies. This program consists of two sides: a cost side and a profit side. On the cost side, eligible firms pay much lower wage taxes on the R&D they perform in the Netherlands. On the profit side, eligible firms face a special corporate income tax regime for the profit generated by the associated R&D. In turn, these incentives lower the effective tax rate that participating firms face.

On the cost side, the program is called WBSO (NL: *Wet Bevordering Speur- en Ontwikkelingswerk*, EN: The Promoting Research and Development Act), and it is administered by the RVO (NL: *Rijksdienst voor Ondernemend Nederland*, EN: the Netherlands Enterprise Agency). The WBSO can be thought of as an R&D remittance reduction of the wage tax and of the contributions to the national insurance scheme. To be eligible for this cost-side benefit, firms need to apply to the RVO for a WBSO statement, showing that (i) they develop a technically new product or production process, write software, or perform scientific research, and (ii) their project contains technical bottlenecks that known techniques cannot solve, which require them to perform R&D.[1] After the RVO assessment, firms receive the WBSO statement and the granted R&D hours, from which they can calculate their "R&D base". With this statement, firms are then eligible for the remittance reduction.

In 2022, this remittance reduction was 32% of the "R&D base" up to 350,000 EUR, and 16% of the R&D base beyond 350,000 EUR. There is also a special rate of 40% up to 350,000 EUR for new entrants, i.e., firms that have employed staff for less than five years and that were granted a WBSO statement in fewer than three calendar years. There are two ways to calculate the R&D base: based on the number of R&D hours of the workers, or based on the actual costs and expenses related to R&D activities.[2]

[1] For example, suppose Firm A submits a project that aims to create a machine learning model using well-known libraries such as TensorFlow. In this case, the RVO will likely not approve the application, since Firm A uses existing technologies to solve its technical issues. If the project instead builds a machine learning model from scratch and, along the way, solves technical bottlenecks, the RVO might grant Firm A the WBSO statement. In short, the main criterion for the RVO approval is whether the project solves technical problems that known methods cannot solve.

[2] For the hours-based method, the firm uses an average hourly wage of 29 EUR for every granted R&D hour. For the first 1,800 granted R&D hours, the firm can add 10 EUR for every hour, and beyond 1,800 hours, 4 EUR for every additional hour.

On the profit side, the program is called the Dutch Innovation Box policy, which provides firms with a reduced corporate income tax rate. The program was introduced in 2007 as the Patent Box policy, and in 2010 it was expanded into the Innovation Box policy (Mohnen et al., 2017). As of 2022, the profit that falls under the policy is taxed at a corporate income tax rate of 9% instead of the standard rate of 25.8%.[3] Note that only the part of the firm's profit that is attributable to R&D is eligible. The main requirement to apply for this policy is the WBSO statement from the RVO. If the firm is considered "large", it also needs a patent or a plant breeders' right (NL: *Kwekersrecht*). A firm is considered "small" if it satisfies two conditions: (i) its gross margin related to the intangible asset over the past five years is less than 37.5 million EUR, and (ii) its turnover over the past five years is less than 250 million EUR.

There are four methods to determine the taxable profit under the Innovation Box policy, and the Dutch Tax Authority uses one of them to determine the profit attributable to R&D. The first and most popular method is the peel-off method (NL: *Afpelmethode*), which attributes parts of the earnings before interest and tax (EBIT) to entrepreneurship, sales, production, and R&D.[4] The second method is the cost-plus method, in which the Tax Authority adds a markup of 8% to 15% to all R&D-related costs. The third method, the single intangible asset method, is similar to the peel-off method, but instead of starting from EBIT, it only uses the R&D costs and the benefits related to a specific intangible asset.[5] If none of these three methods is feasible, the Tax Authority uses the fourth method, the flat-rate method, under which 25% of the profit falls under the Innovation Box.[6]

The Dutch government has adopted the OECD "Nexus Approach" for the Innovation Box Policy, under which only qualifying income relating to intangible assets developed by firms "in-house" will be eligible for the application. This decision effectively prevents firms from benefiting from the tax benefits of the policy if they do not have a substantial economic presence in the Netherlands or if they are not engaged in any research or innovative activities in the country. Furthermore, the decision implies that only innovation developed in-house by firms will be eligible for the tax benefits[7].

[3]Since the expansion in 2010, the rate has risen from 5% to the current 9%. In the Netherlands, there are two corporate income tax brackets. The profit below the 395,000 EUR threshold is taxed at 15%, and the profit beyond 395,000 EUR at 25.8%. For example, if Firm A has a profit of 500,000 EUR in one fiscal year, the first 395,000 EUR is taxed at 15%, and the remaining 105,000 EUR at 25.8%.

[4]The Tax Authority assesses the share of the profit attributable to each part, for example 20% to entrepreneurship, 10% to sales, and 30% to production, which leaves 40% attributable to R&D. The Tax Authority also assesses whether R&D is an essential part of the firm's day-to-day activities, and whether the R&D costs are compensated by the profits attributable to R&D.

[5]The Tax Authority usually uses this method for royalty structures, where it is more straightforward to identify the revenue generated by a specific intangible asset, for example the royalty revenue from a single patent of Firm A.

[6]The benefit is capped at 25,000 EUR per year. The fourth method is usually used for small firms.

[7]Technically, R&D outsourcing to an affiliated entity is feasible as long as the R&D costs and risks are owned by the firm in which it has a significant domestic presence. However, the benefit of doing such a strategy is greatly reduced. Only a maximum of 30% of the R&D expenses can be outsourced to an affiliated entity to be eligible for the Innovation Box benefits. In practice, such eligibility is difficult to attain and, once attained, to demonstrate to the tax authority, so contracted-out R&D receives little effective support under the scheme.

Throughout the paper, I try to capture both the WBSO and the Innovation Box policy in the counterfactual simulations as reductions in the costs of R&D.

2.2 Data

I employ several datasets from various sources. The first dataset is the Production Statistics survey (PS) from Statistics Netherlands (NL: *Centraal Bureau voor de Statistiek*, CBS). This dataset contains information on the firm's characteristics, such as capital, intermediate inputs, and revenue. CBS conducts this survey annually. The survey covers all large firms in the Netherlands, while smaller firms are sampled.[8]

The second dataset is the biennial Community Innovation Survey (CIS). This dataset contains information about the mode of R&D. Let me first discuss the mode of R&D captured in this survey. The definition of the mode of R&D in this paper closely follows the definition outlined by Cassiman and Veugelers (2006), building on Veugelers (1997) and Veugelers and Cassiman (1999). For example, suppose a firm performs in-house R&D with its own staff. In that case, I define this R&D activity as internal R&D. If a firm decides to contract out its R&D to other firms, universities, or other strategic partners, I define this activity as external R&D. Combining the two, each firm in each year falls into one of four R&D modes: no R&D, internal R&D only, external R&D only, or both internal and external R&D. The survey only contains information on the extensive margin, i.e., whether a firm performs a certain type of R&D in a given year.[9]

In this paper, I focus on the Dutch Information, Communication, and Technology (ICT) industry. This industry consists of several sub-industries, such as software publishing, IT services, information services, and telecommunications.[10] I focus on this industry for two reasons. First, the ICT industry is central to Dutch innovation policy, such as the Digital Agenda (Ministerie van Economische Zaken, 2016). Second, the take-up of the WBSO is high in knowledge-intensive sectors such as ICT, with about 1.5 billion EUR in annual benefits economy-wide (Dialogic and SEO Economisch Onderzoek, 2025), and Lokshin and Mohnen (2012) find that the WBSO raises firms' R&D capital. In my dataset, the ICT industry also contributes the largest number of observations, with 750 firms.

I also use deflators for output, capital, and intermediate inputs from the OECD's STAN database. The final dataset is an unbalanced panel of the merged PS and CIS from 2000 to 2020 (even years).

[8] For firms with fewer than ten workers, CBS uses tax registration data as much as possible to fill in the survey. Firms with fewer than 50 workers receive a questionnaire on a sample basis, and all firms with 50 or more workers are included. Details on the survey are available at `https://www.cbs.nl/en-gb/onze-diensten/methods/surveys/korte-onderzoeksbeschrijvingen/production-statistics`.

[9] In the CIS, the firm must fill in the indicator of each type of R&D. However, the amount that the firm spends on each type is not mandatory, and it is not asked in every wave of the survey.

[10] I choose firms that are active in the following two- or three-digit SBI codes (NL: *Standaard Bedrijfsindeling*, EN: Standard Business Classification): 61 (telecommunications), 62 (support activities in the field of information technology), 63 (information service activities), and 582 (software publishing).

Table 1: Firm Characteristics of Dutch ICT Industry

Variables	Average
Output/Revenue	59822.95
Capital	7899.39
Intermediate Inputs	28810.87
Value Added	30353.23
FTE Workers	229.55
Energy	247.57
Share of New Products	11.01%
No R&D	37%
Internal-only R&D	32%
External-only R&D	3%
Both R&D	28%
Observations	2883
Number of Firms	750

Notes.—The unit for firms' output, capital, intermediate inputs, value-added, and energy is thousands of EUR (real). The unit for FTE workers is the number of people employed by a firm.

Table 1 provides information on several variables that I use throughout the paper. Among the 750 firms in the sample (2,883 observations), the average firm has a revenue of about 59.8 million EUR and a capital stock of about 7.9 million EUR. On average, firms use about 28.8 million EUR in intermediate inputs, 230 FTE workers, and 248,000 EUR in energy. As for the R&D modes, 37% of the observations perform no R&D at all, 32% perform only internal R&D, 3% perform only external R&D, and 28% perform both internal and external R&D.

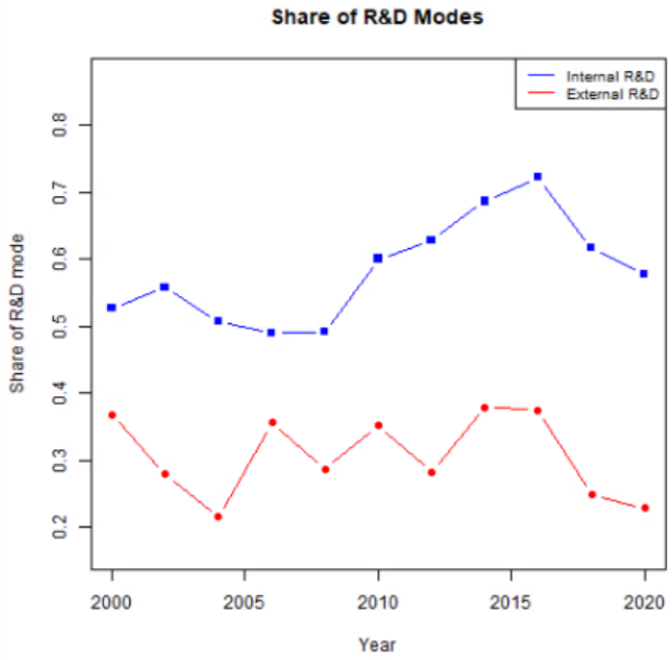
2.3 Motivating Evidence

In this subsection, I provide several facts about the R&D activities of Dutch ICT firms. First, I show the time trends of the R&D modes. Then, I show the persistence of the R&D modes. Finally, I provide reduced-form evidence on whether internal and external R&D are complementary.

Fact 1 (Time-trend of R&D: Internal v. External R&D) *The share of firms performing internal R&D is larger than the share performing external R&D. The share performing internal R&D increases over time, especially after the expansion of the Dutch Innovation Box policy. The share performing external R&D remains roughly constant throughout the sample period.*

Figure 1 shows the time trends of internal and external R&D. The share of firms performing internal R&D is larger than the share performing external R&D throughout the sample period. After the expansion of the Innovation Box policy in 2010, the share performing internal R&D rises from around 50% to a peak of around 70% in 2016. Meanwhile, the share performing external R&D stays between around 20% and 40% throughout the sample period.

Figure 1: Time trend of R&D Activities



Notes.—The figure is constructed from an unbalanced panel of Dutch Production Statistics (PS) and Community Innovation Survey (CIS) datasets by taking the mean of R&D modes in each year. *Internal R&D* indicates that a firm performs Internal R&D in a given year. *External R&D* indicates that a firm performs External R&D in a given year.

Fact 2 (Time-trend of R&D: None, Internal-only, External-only, Both R&D) *The share of firms performing no R&D at all decreases, especially after the expansion of the Dutch Innovation Box policy. The shares of firms performing internal-only R&D and both R&D increase. The share of firms performing external-only R&D is persistently small.*

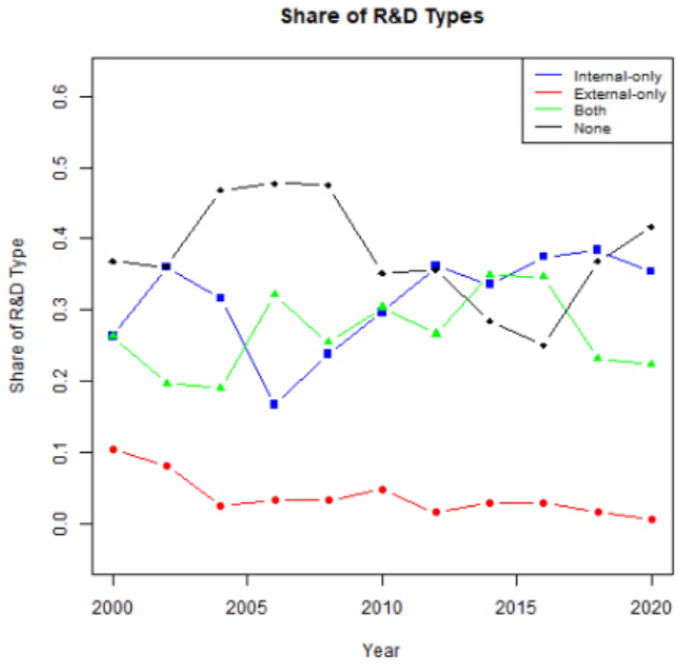
Figure 2 shows the time trends of the four R&D modes. The share of firms performing only internal R&D (the blue line) increases over the years, especially after the expansion of the Innovation Box policy in 2010. The share of firms performing no R&D at all decreases after the expansion, and then increases again in the later years, from 2018 to 2020. As for the share performing both, it rises after 2010 but falls in 2018 and 2020. Meanwhile, the share performing only external R&D is persistently small over the sample period.

Fact 3 (R&D activities persistence) *R&D modes are persistent: a firm's R&D mode in the previous period predicts its R&D mode in the current period.*

Table 2 reports a linear probability model of the current R&D mode on the lagged R&D mode, with year and sub-industry fixed effects.[11] I find that the coefficients are all positive and statistically significant. That is, doing no R&D in the previous period is associated with a 42.02 percentage-point higher probability of doing no R&D in the current period. Similarly, the corresponding numbers are 27.88 percentage points for internal-only, 17.80 for external-only, and 35.38 for both.

[11] A probit specification gives the same sign pattern. See Appendix A.

Figure 2: Time trend of R&D Modes



Notes.—The figure is constructed from an unbalanced panel of Dutch Production Statistics (PS) and Community Innovation Survey (CIS) datasets by taking the mean of R&D modes in each year. *No R&D* indicates that a firm performs no R&D in a given year. *Internal-only R&D* denotes that a firm performs only internal R&D, while *External-only R&D* means that a firm only contracts out R&D to other strategic partners. *Both R&D* marks that a firm conducts both Internal and External R&D.

Table 2: OLS: Persistence of R&D modes

OLS	***Dep. var***			
	No R&D (t)	**Internal-only (t)**	**External-only (t)**	**Both (t)**
No R&D (t-1)	0.4202^{***} (0.0296)			
Internal-only (t-1)		0.2788^{***} (0.0366)		
External-only (t-1)			0.1780^{**} (0.0561)	
Both (t-1)				0.3538^{***} (0.0351)
Year FE	Yes	Yes	Yes	Yes
Sub-Industry FE	Yes	Yes	Yes	Yes

Notes.—OLS of the current period of R&D modes on the previous period of R&D modes. Standard errors are in parentheses. I use heteroskedasticity and autocorrelation-consistent (HAC) standard errors. Significance level: *** : $p < 0.001$, ** : $p < 0.01$, * : $p < 0.05$, $^{\cdot}$: $p < 0.1$.

Fact 4 (Returns to Combining Internal and External R&D) *I find no evidence that internal and external R&D are complementary in outcomes: combining the two modes does not raise value added or the share of new products by significantly more than the two modes separately.*

Table 3: Reduced-form of Complementarity of Internal and External R&D

	Dependent Variable		
	$log(VA)$	**Share New Product**	
	OLS	**OLS**	**Tobit (RE)**
No R&D	9.822^{***}	15.92^{***}	12.81^{***}
	(0.2313)	(3.678)	(2.31)
Internal-only R&D	9.892^{***}	15.36^{**}	25.56^{***}
	(0.2326)	(3.605)	(2.09)
External-only R&D	10.46^{***}	13.82^{**}	18.75^{***}
	(0.3031)	(4.759)	(3.55)
Both R&D	10.33^{***}	18.00^{***}	30.93^{***}
	(0.2360)	(3.612)	(2.14)
Year F.E	Yes	Yes	Yes
Sub-Industry F.E	Yes	Yes	Yes
Wald p-value of $\Delta = 0$	0.38	0.19	0.44

Notes.—Significance level: $^{***}: p < 0.001$, $^{**}: p < 0.01$, $^{*}: p < 0.05$, $^{\cdot}: p < 0.1$. The dependent variable is the share of new products in revenue $\in [0, 100]$. Standard errors are in parentheses. I use heteroskedasticity and autocorrelation-consistent (HAC) standard errors. The last row reports p-values for a Wald test of equal incremental returns, $\Delta = 0$ (Equation (21)). Large p-values mean the extra return to combining the two modes is not distinguishable from adding them separately.

Table 3 provides a simple reduced-form test of complementarity between internal and external R&D. To do so, I follow similar approaches used by Arora (1996), Athey and Stern (1998), and Cassiman and Veugelers (2006), based on the theoretical work of Milgrom and Roberts (1990, 1995).[12] The main intuition is that two activities are complementary if adding one activity while the other is already performed raises the outcome by more than adding it in isolation. That is, complementarity requires $\theta_{\text{both}} - \theta_{\text{internal}} \geq \theta_{\text{external}} - \theta_{\text{no}}$ (Equation (21)). I find that the four coefficients are positive and significant in all three specifications. However, the last row of the table shows that I cannot reject the null hypothesis that the two increments are equal, i.e., $\Delta = 0$, where Δ is the difference between the two increments. The Wald p-values are 0.38, 0.19, and 0.44. Furthermore, the point estimates satisfy the inequality only for the share of new products in the OLS specification, while the value-added and Tobit point estimates go the other way. As such, if the two modes are complementary, the complementarity does not show up in outcomes. In the model, I instead look for it in costs, from how often firms choose to do both (Section 4).

Discussion on Identification. However, the above reduced-form results come with important caveats. First, running a pooled OLS while ignoring the censored part of the data gives biased and inconsistent estimates, since the censored sample is not representative of the population.[13] I

[12] See Appendix B for details.

[13] There are more disturbances above the true regression line. In this case, the distribution is truncated, while OLS

address this issue by running a panel random-effects Tobit model. Second, even though I reduce the bias by using the Tobit model and exploiting the panel structure of the data, the endogeneity problem is not resolved, and adding controls does not resolve it either. Finally, as pointed out by Athey and Stern (1998), there is still a risk that these R&D activities are driven by unobserved heterogeneity. As such, I interpret these results with caution. These caveats are my primary motivation to develop the dynamic model in the next section.

3 Model

In this section, I present my model. My focus here is to understand how firms decide which mode of R&D to perform. The model has two parts: a static part and a dynamic part. In the static part, I discuss the supply and demand sides of the model, and the firm's revenue and profit under monopolistic competition. In the dynamic part, I outline a single-agent dynamic discrete choice over four R&D modes, in the tradition of Rust (1987). As in Igami (2017) and Igami and Uetake (2020), each mode has its own investment cost, and, as in Miravete and Pernias (2006), a cost-complementarity parameter captures whether doing both internal and external R&D is cheaper than the sum of the two costs. Finally, I interpret the cost of external R&D and the cost-complementarity through a hold-up problem between the firm and its external R&D partner (Grossman and Hart, 1986).

3.1 Static Decisions: Supply and Demand

The general structure of the static part follows Aw et al. (2011), Boler et al. (2015), and Peters et al. (2017)[14]. In particular, I assume monopolistic competition in the firm's static decisions. Monopolistic competition approximates a market in which strategic interactions among firms are weak (see Zhelobodko et al. (2012) and Thisse and Ushchev (2018) for the discussion of the general case of monopolistic competition).

Demand. The representative consumer has a CES-like utility function. By implication, the demand curve faced by firm i is then assumed to follow the Dixit-Stiglitz form. That is,

$$Q_{it} = Q_t \left(\frac{P_{it}}{P_t} \right)^{\eta} \tag{1}$$

where P_{it} is the firm's output price, P_t is the average price in the period t, and η is the elasticity of demand, which is assumed to be constant for all firms in the industry.

places equal weights on positive and negative disturbances.

[14] ***Alternative Demand System.***—One might argue that the demand side of my model is not as microfounded as in the standard empirical IO literature, since I use a CES-like utility function. As shown by Anderson et al. (1992) and De Loecker (2011), this demand system can be generated from a logit model of consumer choice with log prices, which yields a similar revenue equation (see Appendix E). The main difference from the models of Berry (1994) and Berry et al. (1995) is that the own-price and cross-price elasticities is the same by assumption. I use the CES-like demand because I do not observe prices and quantities at the firm level, which precludes estimating a demand system as in Berry et al. (1995).

Supply. Now, suppose firm i's (log of) marginal cost is defined as follows

$$c_{it} = c(k_{it}, l_{it}, w_t) - \omega_{it} = \beta_0 + \beta_k k_{it} + \beta_l l_{it} + \beta_w ln(w_t) - \omega_{it} \tag{2}$$

where k_{it} is the log of firm's capital stock, l_{it} is the log of the number of FTE workers, w_t is a vector of variable input prices common to all firms, and ω_{it} is the firm's productivity. The firm is assumed to produce a single output. There are several sources of cost heterogeneity in this model[15]. The first one is coming from the firm's capital stock which is directly observable in the data. The second source is from the number of workers the firms have, which is also directly observable in the data. The third source is the firm's productivity which is observed by the firm, but not observable by the econometrician. Notice here that the more productive the firm is, the lower the marginal cost it faces. The marginal cost does not vary with the firm's output level, implying that demand shocks in one market do not affect the static output decision in other markets. Therefore, I can compute revenue and profits in each market independently of the output levels in other markets.

Revenue. Using information from the demand-side and the marginal cost, I can get the log of firm's revenue as follows

Result 1 (Revenue equation) *The (log) revenue equation is derived from both the demand and supply sides, and follows the following expression*

$$r_{it} = (\eta + 1) ln\left(\frac{\eta}{\eta + 1}\right) + ln(\Phi_t) + (\eta + 1)\left[\beta_0 + \beta_k k_{it} + \beta_l l_{it} + \beta_w ln(w_t) - \omega_{it}\right] \tag{3}$$

where $\Phi_t := Q_t / P_t^{\eta}$.

Proof. See Appendix C for the full derivation. ■

From the above expression, the firm's revenue provides information on its marginal cost, in particular the productivity level ω_{it}. In the empirical implementation, I estimate the revenue function, which allows me to recover ω_{it} as the part of the firm's marginal cost that capital, labor, and common input prices do not explain. Note that while I describe ω_{it} as the firm's productivity, it may also capture the quality of the firm's product, since I observe revenue and not prices or quantities.

Static Profit. I can now recover the profit of the firm using the expression of the firm's revenue above as follows

Result 2 (Static profit equation) *The static profit equation is derived from both the revenue equation and the (log) marginal cost equation, and follows the following expression*

[15] ***Alternative Supply-Side*** .—The lack of information on prices, the number of products each firm produces, and the quality of each product precludes an analysis of the supply-side as in the standard empirical IO literature. For example, the absence of firm's output price makes it hard to model the supply-side as an oligopoly. In this model, I also assume a single-product firm throughout the economy since I do not observe the number of products.

$$\pi_{it} = -\left(\frac{1}{\eta}\right) exp(r_{it}(k_{it}, l_{it}, \omega_{it}; \eta, \beta_0, \beta_k, \beta_l)) \tag{4}$$

Proof. The revenue expression follows from Result 1 by taking the exponent. The cost expression follows from taking the exponent of the log marginal cost with the quantity from the demand side. To retrieve the static profit, I simply subtract cost from revenue. See Appendix C for the full derivation. ■

The above expression allows me to recover the profit of the firm using estimates of the model and observables in the data. Note that, this profit will be an important ingredient of the firm's decision to choose modes of R&D in the dynamic discrete choice model developed in the next subsections.

3.2 Dynamic Decisions: Dynamic Discrete Choice of R&D modes

As mentioned in Section 2, I try to capture the WBSO through some form of subsidy from the government. In particular, if the firm receives this subsidy, it experiences a decrease in its costs of investing in certain modes of R&D. As such, the following single-agent dynamic discrete choice model will incorporate investment costs as its main elements.[16]

Setup. I assume that time is discrete with infinite horizon. A finite number of firms are indexed by i. The firm's state space is defined as $s_{it} = (\omega_{it}, k_{it}, l_{it})$, which endogenously evolves as the firm decides which mode of R&D to choose. ω_{it} is the (discretized) productivity of firm i in a year t. In each year, firm i can choose to do no R&D at all, only internal R&D, only external R&D, or both internal and external R&D.

Each mode of R&D entails specific investment costs. For example, suppose firm i decides to perform internal R&D. In that case, it needs to invest a certain cost of $\kappa^{internal}$ to procure the internal R&D. If the firm decides to do external R&D, then, it needs to invest a certain cost of $\kappa^{external}$ in order to perform the external R&D. If the firm decides to do both R&D, it has to pay both types of investment costs. These costs are mode-specific investments that firms need to pay to perform a particular R&D. For firms that choose not to do R&D at all, there is no specific investment cost.

There are several possible reasons why a firm decides to do both internal and external R&D. First, both types of R&D may benefit the firm in terms of cost savings or better product quality. For example, in the Dutch ICT industry, a firm focusing on online booking technology and services

[16]Since the model is a single-agent problem, a firm's profit and productivity depend only on its own R&D mode. Strategic interactions through R&D are then absorbed into the estimated costs. For example, competing with rivals for the same contractor or university lab would show up as a higher cost of external R&D, and waiting to see a rival's product would show up as choosing no R&D. Separating these interactions from the costs of each mode would require firm-level price data, which I do not observe.

can decide to build a dynamic pricing algorithm from scratch (i.e., an internal R&D). At the same time, this firm can also consult with economists and computer scientists at several universities on the technical know-how on issues and state-of-the-art solutions related to the dynamic pricing algorithm (i.e., an external R&D). Doing both types of R&D allows the firm to gain an additional benefit from faster technological adoption of the dynamic pricing algorithm, even though it incurs an additional investment cost of doing both R&D types. I call this phenomenon the cost-complementarity of internal and external R&D. It is also possible that doing both internal and external R&D yields no meaningful benefit for the firm. Following Miravete and Pernias (2006), I capture the cost-complementarity of internal and external R&D in a parameter $\kappa^{complement}$. In particular, I assume that this parameter $\kappa^{complement}$ can take on the following values.

$$\kappa^{complement} \begin{cases} > 0, & [Complement] \\ = 0, & [Independent] \\ < 0, & [Substitute] \end{cases}$$

In the above expression, three possible values exist for the cost-complementarity parameter, $\kappa^{complement}$. If it is positive, then both types of R&D complement each other. It is also possible that doing both types of R&D does not benefit the firm, i.e., both are independent. I also include in the parameter $\kappa^{complement}$ the possibility that both types of R&D are substitutes.

Each firm draws an IID private cost shock $\epsilon_{it}^{a_{it}}$ which follows a Type-1 Extreme Value distribution. After observing the four shocks, firm i takes an action a_{it} from the choice set $a_{it} \in \{\text{none}, \text{internal-only}, \text{external-only}, \text{both}\}$. Private shocks reflect each firm's informational, managerial, and organizational conditions. If the firm chooses a mode that involves external R&D, it then invests in the relationship with its partner and bargains over the value of the collaboration. That is, the firm first chooses its R&D mode, and only then bargains with its partner. This timing matters for the choice probabilities. Because the four shocks are drawn before any bargaining takes place, the choice probabilities keep the logit form that I derive below.

State Transitions. I assume that ω_{it} follows a controlled Markov process. In particular, the distribution of ω_{it} can be written as $G_\omega(\omega_{it} \mid \mathcal{I}_{it-1}) = G_\omega(\omega_{it} \mid \omega_{it-1}, \text{none}_{it-1}, \text{int}_{it-1}, \text{ext}_{it-1}, \text{both}_{it-1})$, where $\mathcal{I}_{it}$ is an information set which contains the information that the firm can use to solve its decision problem in period t. I can express the persistent productivity ω_{it} as $\omega_{it} = g_\omega(\omega_{it-1}, \text{none}_{it-1}, \text{int}_{it-1}, \text{ext}_{it-1}, \text{b}$ ξ_{it}, where $E[\xi_{it} \mid \mathcal{I}_{it-1}] = 0$. Variables $\text{none}_{it-1}, \text{int}_{it-1}, \text{ext}_{it-1}, \text{both}_{it-1}$ are indicator variables (in period $t-1$) that refer to the decision of the firm to choose no R&D at all, only internal R&D, only external R&D, and both R&D, respectively. The random variable ξ_{it} captures the (unanticipated at period $t-1$) stochastic nature of improvement on the firm's persistent productivity ω_{it} in period t. By construction, ξ_{it} does not correlate with ω_{it-1}. Note that $\text{none}_{it-1}, \text{int}_{it-1}, \text{ext}_{it-1}, \text{both}_{it-1}$ are assumed to be predetermined variables; i.e., these variables are realized before ω_{it} is realized and functions of previous period information set; that is, $\{\text{none}_{it-1}, \text{int}_{it-1}, \text{ext}_{it-1}, \text{both}_{it-1}\} =$

$\mathbb{X}(\mathcal{I}_{it-1}) \in \mathcal{I}_{it}$. Therefore, those variables also do not correlate with ξ_{it}. Note that the transition depends only on which of the four modes the firm chose in the previous period. In particular, how much the firm invests in the relationship with its external partner does not enter the transition.

Dynamic Programming Problem. Firms make their dynamic discrete choices to maximize their expected values. The future stream of profits is discounted by a factor $\beta \in (0,1)$. The following Bellman equation characterizes the dynamic programming problem of each firm:

$$
\begin{aligned}
V(s_{it}) = \pi_{it}(s_{it}) \\
&+ \max \Bigg\{ \underbrace{\beta EV^{\text{no R\&D}}(s_{it}) + \epsilon_{it}^{\text{no R\&D}}}_{\text{No R\&D}}; \\
&\underbrace{-\kappa^{\text{internal}} + \beta EV^{\text{internal}}(s_{it}) + \epsilon_{it}^{\text{internal}}}_{\text{Internal-only}}; \\
&\underbrace{-\kappa^{\text{external}} + \beta EV^{\text{external}}(s_{it}) + \epsilon_{it}^{\text{external}}}_{\text{External-only}}; \\
&\underbrace{-\kappa^{\text{internal}} - \kappa^{\text{external}} + \kappa^{\text{complement}} + \beta EV^{\text{both}}(s_{it}) + \epsilon_{it}^{\text{both}}}_{\text{Both}} \Bigg\}
\end{aligned}
\tag{5}
$$

As explained, the key parameters of this dynamic discrete choice R&D model are the investment cost of internal R&D ($\kappa^{internal}$), the investment cost of external R&D ($\kappa^{external}$), and the cost-complementarity parameter ($\kappa^{complement}$). Here, I assume that unobservables ϵ_{it}^{a} are additive to the key parameters and the expected future value, $EV(\cdot)$. I also assume that the scale parameter is normalized to one, $\sigma = 1$.

To illustrate the Bellman equation (5), consider again the Dutch ICT firm that develops a dynamic pricing algorithm. In each year, the firm compares four options. First, if the firm does no R&D, it pays no R&D cost, and its productivity evolves as for a firm without R&D. Second, if the firm builds the algorithm in-house, it pays $\kappa^{internal}$, and its productivity evolves as for a firm with internal R&D. Third, if the firm only consults university researchers, it pays $\kappa^{external}$ instead. Finally, if the firm does both, it pays both costs, but it saves $\kappa^{complement}$. Each option also carries its own private shock. As such, the firm may choose to do both even when doing both costs more than doing internal R&D alone. This happens when the expected future value from doing both, together with its private shock, is high enough to compensate for the extra cost.

I follow Rust (1987, 1994) in exploiting the property of the logit error and the assumption of the conditional independence over time. I get a closed-form expression for the expected value *before* observing private shocks $\epsilon_{it}^{a_{it}}$ as follows (i.e., the McFadden's social surplus)

Result 3 (McFadden's Social Surplus) *The McFadden's social surplus can be expressed as follows*

$$\begin{aligned} EV(s_{it}) = \pi_{it}(s_{it}) + \gamma^{Euler} \\ + ln\left[exp(\overline{V}_{it}^{no}) + exp(\overline{V}_{it}^{internal}) + exp(\overline{V}_{it}^{external}) + exp(\overline{V}_{it}^{both})\right] \end{aligned} \tag{6}$$

where γ^{Euler} is the Euler's constant. $\overline{V}_{it}^{no}, \overline{V}_{it}^{internal}, \overline{V}_{it}^{external}, \overline{V}_{it}^{both}$ represent the deterministic part of the conditional (or "alternative-specific") values in Equation (5)*.*

Proof. The derivation of McFadden's social surplus is in the Appendix F ■

The main advantage of this expression is that I do not have to integrate over the distribution of the four shocks in each state. Instead, the expected value before the firm observes its shocks has a closed form. I use this closed form to solve the Bellman equation, to derive the choice probabilities, and to estimate the model.

As for the expected future value $(EV(\cdot))$, I can define it as follows

Result 4 (Expected future value) *The expected future value can be expressed as follows*

$$\begin{aligned} EV^{a}(s_{it}) = \sum_{\omega'} V(s_{it+1})\, P(\omega' \mid s_{it}, a_{it}) \\ , \qquad \forall a_{it} \in \{no\ R\&D, internal, external, both\} \end{aligned} \tag{7}$$

where $P(\omega' \mid s_{it}, a_{it})$ can be thought of as the transition probability or, equivalently, the state transition.

Proof. The derivation of $EV(\cdot)$ can be found in the Appendix G. ■

I can now define the conditional choice probabilities as follows

Result 5 (Conditional choice probabilities) *The conditional choice probabilities can be expressed as follows*

$$\begin{aligned} Pr(a_{it} = action) = \frac{exp(\overline{V}_{it}^{action})}{exp(\overline{V}_{it}^{no}) + exp(\overline{V}_{it}^{in}) + exp(\overline{V}_{it}^{out}) + exp(\overline{V}_{it}^{both})} \\ , \qquad \forall a_{it} \in \{no\ R\&D, internal, external, both\} \end{aligned} \tag{8}$$

Proof. The full derivation of the probability can be found in the Appendix H. ■

I use these optimal choice probabilities to construct a likelihood function for the empirical implementation.

3.3 Hold-up and the Cost of External R&D

In this subsection, I interpret two of the three costs in the Bellman equation (5), the cost of external R&D and the cost-complementarity, through a hold-up problem on the external R&D contract

(Grossman and Hart, 1986; Hart and Moore, 1990), in the tradition of Aghion and Tirole (1994) on the organization of research.[17] To fix ideas, consider again the Dutch ICT firm that develops a dynamic pricing algorithm, and suppose it consults university researchers on the remaining technical bottlenecks. To make the collaboration work, the firm has to invest in it, for example by preparing its booking data and explaining its pricing problem to the researchers. Once the firm has made this investment, the investment has little value outside the collaboration, and the contract cannot specify in advance how the value of the results will be shared. The researchers can then capture part of that value. Anticipating this, the firm invests too little in the collaboration, and contracting out becomes costly. If the firm also has its own R&D staff working on the algorithm, its position in the collaboration is stronger. The firm can specify what it needs, evaluate what the researchers deliver, and, if needed, continue the work in-house. As a result, the firm keeps a larger share of the value, and contracting out becomes cheaper. In this model, in-house R&D acts on the share of the value that the firm keeps from the collaboration, while its effect on productivity is captured separately, through the state transition above. This idea is related to the absorptive-capacity argument of Cohen and Levinthal (1989, 1990), in which in-house R&D helps the firm make use of external knowledge.

Formally, consider a choice that involves external R&D, $a_{it} \in \{\text{external-only}, \text{both}\}$. The firm makes a relationship-specific investment $I \geq 0$ at a cost of I^2. This investment raises the value of the collaboration by θI, where $\theta > 0$ is a surplus scale measured in the same units as the Bellman equation (million EUR). Because the contract is incomplete, the partner can hold the firm up once I is sunk. If the two parties disagree at that point, both receive zero, i.e., the investment has no value outside this collaboration. The firm then keeps a share $\tau \in (0, 1]$ of the value, and the partner keeps the remaining $1 - \tau$. When the firm also conducts internal R&D, its share rises to $\tau' = \tau + \delta$, where $\delta \geq 0$ measures how much in-house R&D strengthens the firm's position. With zero disagreement payoffs and a transferable value, this split is both the asymmetric Nash bargaining solution (Nash, 1950) and the proportional solution of Kalai (1977). I treat τ as a primitive of the model.

Anticipating the split, the firm chooses its investment to solve

$$\max_{I \geq 0} \ \tau\theta I - I^2. \tag{9}$$

Result 6 (Equilibrium net payoff) *The unique solution of* (9) *gives the firm's equilibrium net payoff*

$$\text{net}(\tau) = \frac{\tau^2\theta^2}{4}. \tag{10}$$

[17] Other models can produce the same pattern of costs. For example, haggling over an incomplete contract (Bajari and Tadelis, 2001; Tadelis, 2002) also makes external R&D costly, and in-house R&D also makes the external contract cheaper to use. Since the data record only the R&D modes and not the contracts themselves, I cannot distinguish these explanations. I use the hold-up problem because it has a closed-form solution that fits directly into the four-mode problem.

If the firm kept the entire value of the collaboration, it would solve the same problem with $\tau = 1$. I define the resulting surplus as

$$S := \text{net}(1) = \frac{\theta^2}{4}. \tag{11}$$

Then $\text{net}(\tau) = S\tau^2$. Measuring the firm's net payoff against this surplus gives two of the costs in the Bellman equation:

$$\kappa^{external} = \text{net}(1) - \text{net}(\tau) = S\big(1 - \tau^2\big), \tag{12}$$

$$\kappa^{complement} = \text{net}(\tau') - \text{net}(\tau) = S\big((\tau')^2 - \tau^2\big). \tag{13}$$

Both costs are therefore functions of S, τ, and τ'.

Proof. See Appendix J for the first-order condition, the second-order condition, and the location identities. ■

The intuition of Result 6 is the following. Think of τ as the share of the value that the firm expects to keep when it contracts out without its own R&D staff. The smaller this share, the less the firm invests in the collaboration, the smaller its net payoff, and the larger the cost of external R&D. To see the magnitudes, take a firm that keeps half of the value, $\tau = 0.5$. Its net payoff is then a quarter of the surplus, $\text{net}(0.5) = 0.25S$, so the cost of external R&D is $0.75S$. Two forces make up this cost. First, the firm gives half of the value of the collaboration to its partner, which amounts to $0.5S$. Second, because the firm keeps only half of the value, it invests only half of the efficient amount, so the collaboration creates less value. This underinvestment costs another $0.25S$. Now suppose the same firm also has its own R&D staff, and its share rises to $\tau' = 0.9$. Its net payoff rises to $0.81S$, so the cost-complementarity is $0.81S - 0.25S = 0.56S$. That is, in-house R&D offsets $0.56/0.75 \approx 75\%$ of the extra cost of contracting out.

Result 7 (Payoffs in the Bellman equation) *The current-period payoffs in* (5) *are the firm's equilibrium net payoffs, measured against S,*

$$v^{external} = \text{net}(\tau) - \text{net}(1) = -\kappa^{external}, \tag{14}$$

$$v^{both} = -\kappa^{internal} + \text{net}(\tau') - \text{net}(1) = -\kappa^{internal} - \kappa^{external} + \kappa^{complement}. \tag{15}$$

Proof. See Appendix J. ■

Result 7 shows that the current-period payoffs of external-only and both in the Bellman equation are the firm's net payoffs from the collaboration, measured against the surplus it would collect if it owned the collaboration outright. As such, the cost of external R&D is the value that the firm loses by contracting out without its own R&D staff, which consists of the value transferred to the partner and the value lost to underinvestment. Similarly, the cost-complementarity is the part of that loss that the firm recovers when it also does in-house R&D. Note that the in-house R&D itself still costs $\kappa^{internal}$.

Since the bargaining outcome does not enter the productivity transition, and the firm bargains only after it has chosen its mode, the hold-up model gives the same choice probabilities as the Bellman equation (5). As such, the data identify the three costs, but not the bargaining shares and the surplus scale S separately. Given a value of S, the two costs can be inverted into bargaining shares, which I do in Appendix J.4. However, one object does not require a value of S: the ratio $\kappa^{complement}/\kappa^{external} = \left((\tau')^2 - \tau^2\right)/\left(1-\tau^2\right)$, i.e., the share of the extra cost of contracting out that is offset when the firm also does in-house R&D. In the example above, this share is approximately 75%. Table 6 reports it for the estimated costs.

4 Empirical Implementation

In the previous section, I presented a model with several components that need to be estimated. I estimate them in three steps. First, from the static part of the model, I retrieve the elasticity of demand, η, from the ratio of total variable cost to revenue under monopolistic competition. Second, I estimate the firm's productivity, ω_{it}, together with the parameters that govern how productivity evolves with the firm's R&D mode. The demand elasticity and productivity allow me to compute the static profit of each firm, $\pi_{it}(s_{it})$, and the productivity parameters give me the transition probabilities. Finally, using these estimates, I estimate the three dynamic costs of the model: the cost of internal R&D ($\kappa^{internal}$), the cost of external R&D ($\kappa^{external}$), and the cost-complementarity ($\kappa^{complement}$).

4.1 Retrieving Elasticity of Demand

First, let me recover the elasticity of demand, η. Under monopolistic competition, the firm sets its price as a constant markup over its marginal cost. As a result, the ratio of the firm's total variable cost to its revenue equals $1 + 1/\eta$.

Result 8 (Elasticity of demand) *Under monopolistic competition, the ratio of total variable cost (TVC) to revenue (R) is*

$$\frac{TVC}{R} = 1 + \frac{1}{\eta}$$

Proof. From the static part of the model, revenue is $R_{it} = P_{it}Q_{it}$, and total variable cost is $\frac{1+\eta}{\eta}R_{it}$. Taking the ratio of the two gives the expression above. ■

I use the mean of the cost-revenue ratio across firms in the Dutch ICT industry as an estimate of one plus the inverse of the demand elasticity. I define the total variable cost as the sum of the nominal intermediate inputs and the nominal energy input that the firm uses. As for the revenue, I use the nominal gross output. After retrieving this ratio, it is straightforward to get the demand elasticity η.

Source of identification.—The demand elasticity is identified from the markup implied by monopolistic competition. Since both the total variable cost and the revenue of each firm are observed in the data, I can retrieve η directly. Alternatively, as in Aw et al. (2011), Boler et al. (2015), and Peters et al. (2017), one can estimate η from a regression of total variable cost on revenue with fixed effects:

$$TVC_{it} = R_{it}\left(1 + \frac{1}{\eta}\right) + \gamma_j + \gamma_t + \varepsilon_{it},$$

where γ_j and γ_t are sub-industry and time fixed effects.

4.2 Productivity Estimation

Next, to recover productivity ω_{it}, I use a semiparametric approach similar to Olley and Pakes (1996), Aw et al. (2011), Doraszelski and Jaumandreu (2013), and Peters et al. (2017). The main concern is that the firm's inputs are correlated with its productivity, which the econometrician does not observe. I address this concern in two stages. In the first stage, I use the firm's intermediate inputs, m_{it}, and energy, n_{it}, to control for the unobserved productivity in the revenue equation. The idea of Olley and Pakes (1996) is that the firm chooses these flexible inputs after it observes its productivity, so the inputs contain information on its productivity level. From this first stage, I obtain a fitted value $\hat{\phi}_{it}$, which combines the effects of capital, labor, and productivity, scaled by the demand elasticity. In the second stage, I use the assumption from Section 3 that productivity follows a controlled Markov process, which depends on lagged productivity and the lagged R&D mode. Substituting $\hat{\phi}_{it}$ into that process gives the following estimating equation.

Result 9 (Estimating equation of productivity) *The estimating equation to retrieve productivity ω_{it} takes the following form:*

$$\begin{aligned}\hat{\phi}_{it} = {} & \beta_k^* k_{it} + \beta_l^* l_{it} + \alpha_1\left(\hat{\phi}_{it-1} - \beta_k^* k_{it-1} - \beta_l^* l_{it-1}\right) \\ & - \alpha_2^*\left(\hat{\phi}_{it-1} - \beta_k^* k_{it-1} - \beta_l^* l_{it-1}\right)^2 \\ & + \alpha_3^*\left(\hat{\phi}_{it-1} - \beta_k^* k_{it-1} - \beta_l^* l_{it-1}\right)^3 \\ & - \alpha_5^* no_{it-1} - \alpha_6^* internal_{it-1} - \alpha_7^* external_{it-1} - \alpha_8^* both_{it-1} - \xi_{it}^*\end{aligned} \tag{16}$$

where $\xi_{it}^* \sim \mathcal{N}(0, \sigma_{\xi^*})$, and the star $(*)$ indicates that a coefficient is scaled by the demand elasticity (see Appendix D). I estimate this equation by nonlinear least squares.

Proof. See Appendix D. ■

Using Results 8 and 9, I can then construct an estimate of productivity for each observation as follows

Result 10 (Estimated productivity) *The estimate of productivity can be expressed as the following*

$$\hat{\omega}_{it} = -(1/(\hat{\eta}+1))\hat{\phi}_{it} + \hat{\beta}_k k_{it} + \hat{\beta}_l l_{it} \tag{17}$$

Proof. See Appendix D. ■

Source of identification.—This approach is essentially a semiparametric one. As noted by Doraszelski and Jaumandreu (2013), the main ingredient for identification is that the variables in the parametric part of the model are not perfectly predictable by the variables in the nonparametric part. The first stage also relies on two standard assumptions of the control function literature (Olley and Pakes, 1996; Levinsohn and Petrin, 2003; Ackerberg et al., 2015; Gandhi et al., 2020). The first one is the scalar unobservability assumption, i.e., productivity is the only unobservable that drives the firm's choice of intermediate inputs and energy. The second one is that the demand for these inputs is strictly monotone in productivity, which allows me to invert it and to write productivity as a function of observables. As for the timing, I assume that capital and labor are predetermined, i.e., chosen before ω_{it} is realized, while intermediate inputs and energy are flexible, i.e., chosen after ω_{it} is realized. Finally, the estimating equation also uses the demand elasticity, whose identification I discussed in the previous subsection.

4.3 Costs of R&D and Complementarity

Finally, I estimate the three dynamic costs, $\kappa = (\kappa^{internal}, \kappa^{external}, \kappa^{complement})$. In the previous section, I derived the conditional choice probabilities of the four R&D modes in Equation (8). I use these probabilities to estimate the costs by maximum likelihood, where the likelihood function for the firms' R&D modes is defined as follows

$$L(a_{it}|s_{it};\kappa) = \prod_{i}^{N}\prod_{t}^{T_i} Pr(a_{it} = action)^{I_{\{a_{it}=action\}}} \tag{18}$$

There are two loops in estimating the dynamic discrete choice model. The maximum likelihood estimation serves as the outer loop in estimating Equation (18). Within each loop of the MLE, there is an inner loop involving the calculation of value functions. Essentially, it is a contraction mapping of the value function described in Equation (6). The algorithm to recover the parameters follows the successive-approximations variant of the nested fixed-point algorithm (NFXP-SA), as in Rust (1987)[18].

Source of identification.—Before discussing the identification of the costs, note that the dynamic part of my model is a single-agent dynamic model rather than a fully-fledged dynamic game. The firm's static profit and its transition probabilities do not depend on the actions of other firms,

[18]There are several ways to improve the speed of NFXP-SA; the first one is to perform mathematical programming with equilibrium constraints (MPEC), developed by Su and Judd (2012). Another approach is to change the successive approach algorithm to the Newton-Kantorovich algorithm (NFXP-NK) developed by Iskhakov et al. (2016).

which is an implication of monopolistic competition in the static part of the model. For the dynamic part, I rely on the standard assumptions for the identification of single-agent dynamic models (Rust, 1987, 1994; Aguirregabiria et al., 2021). The private shocks are additive and follow a Type-1 extreme-value distribution with scale normalized to one. The conditional independence assumption holds, i.e., the realization of productivity is independent of the private shocks. The current-period payoff of no R&D is normalized to zero. I do not estimate the discount factor, since its identification is known to be problematic (Rust, 1994; Magnac and Thesmar, 2002; Abbring and Daljord, 2020), and I set it to $\beta = 0.95$.

Given these assumptions, the three costs are identified from how often firms choose each R&D mode. Appendix I writes the three logit inversions, following the conditional-choice-probability map of Hotz and Miller (1993). To see the intuition, take the shares of the four modes in the sample: 36.8% of the observations choose no R&D, 32.4% internal-only, 3.2% external-only, and 27.7% both. As I show in Section 5, the productivity gains are similar across modes, so the expected future values differ little across modes, and the choice frequencies mainly reflect the current costs. If the expected future values were the same, the cost of external R&D would be about $\log(36.8/3.2) \approx 2.44$ million EUR. That is, the very small share of firms that choose external-only identifies a high cost of external R&D. Similarly, the cost of internal R&D would be about $\log(36.8/32.4) \approx 0.13$ million EUR, and the net cost of doing both about $\log(36.8/27.7) \approx 0.28$ million EUR. Since the net cost of doing both is much smaller than the sum of the two separate costs, the large share of firms that choose both identifies a positive cost-complementarity. Finally, as discussed in Section 3.3, the hold-up interpretation leaves the likelihood unchanged.

5 Results

In this section, I first present the estimated elasticity of demand and the productivity parameters from the static part of the model. I then present the estimated costs of internal and external R&D and the cost-complementarity, and what they imply for the hold-up interpretation.

5.1 Demand Elasticity and Productivity

As discussed in Subsection 4.1, I compute the ratio of total variable cost to revenue, as implied by the model. I find that the elasticity of demand for the Dutch ICT industry is $\hat{\eta} = -1.794$, with a standard error of 0.164, which I compute using the delta method. That is, a 1% increase in the firm's price lowers the quantity demanded by about 1.8%. Equivalently, the total variable cost of the firm is on average about 44% of its revenue. Aw et al. (2011) and Peters et al. (2017) estimate demand elasticities with a similar approach for other industries. I use this elasticity in the productivity estimation and in the construction of the firm's static profit.

Next, I discuss the productivity parameters. Table 4 reports the estimates of Result 9, which I obtain by nonlinear least squares. First, I find that productivity is persistent: the coefficient on

lagged productivity is $\hat{\alpha}_1 = 0.704$ and statistically significant. Second, I find that all four R&D mode parameters are positive and statistically significant, and that they are close to each other: $\hat{\alpha}_5 = 0.345$ for no R&D, $\hat{\alpha}_6 = 0.384$ for internal-only, $\hat{\alpha}_7 = 0.390$ for external-only, and $\hat{\alpha}_8 = 0.387$ for both.

Table 4: Estimated Productivity Parameters

Parameters	**Productivity Estimation Dependent var: ϕ**	**Parameters**	**Productivity Estimation Dependent var: ϕ**
$\beta_k(k)$	-0.193^{***} (0.012)	$\alpha_5(no_rd)$	0.345^{***} (0.084)
$\beta_l(l)$	-0.650^{***} (0.025)	$\alpha_6(internal)$	0.384^{***} (0.084)
$\alpha_1(\omega)$	0.704^{***} (0.077)	$\alpha_7(external)$	0.390^{***} (0.102)
$\alpha_2(\omega^2)$	0.051^{*} (0.024)	$\alpha_8(both)$	0.387^{***} (0.085)
$\alpha_3(\omega^3)$	-0.002 (0.002)		
Subindustry FE	Yes		
Year FE	Yes		
$SD(\xi)$	0.347		
Wald p-value of $\Delta = 0$	0.54		

Notes.—Significance level: $^{***}: p < 0.001$, $^{**}: p < 0.01$, $^{*}: p < 0.05$, $^{\cdot}: p < 0.1$. Standard errors are in parentheses. I use heteroskedasticity and autocorrelation-consistent (HAC) standard errors. $SD(\xi)$ indicates the standard deviation of the residuals from the estimating Equation (36). The last row is a Wald p-value of equal incremental returns on $\alpha_5, \ldots, \alpha_8$.

To test whether combining the two modes adds an extra return, I compute the difference $(\alpha_8 - \alpha_6) - (\alpha_7 - \alpha_5)$, i.e., the return to adding external R&D when the firm already does internal R&D, compared with the return to adding external R&D in isolation. This difference is -0.042, and a Wald test of the null hypothesis that it is zero has a p-value of 0.54. That is, I cannot reject that the return to R&D is the same whether or not the firm combines the two modes. To translate the mode parameters into productivity, recall that Equation (17) recovers productivity as $\hat{\omega}_{it} = -(1/(\hat{\eta}+1))\hat{\phi}_{it} + \hat{\beta}_k k_{it} + \hat{\beta}_l l_{it}$, so a difference in the mode parameters maps into ω by the factor $-1/(\eta+1)$. Given $\hat{\eta} = -1.794$, moving from no R&D raises productivity by $(\alpha_5 - \alpha_6)/(\eta+1) \approx 0.049$ for internal-only, by $(\alpha_5 - \alpha_7)/(\eta+1) \approx 0.057$ for external-only, and by $(\alpha_5 - \alpha_8)/(\eta+1) \approx 0.053$ for both. Since ω is in logs, those shifts are about 5.0%, 5.8%, and 5.4%, respectively, which is in the range of returns to R&D found in the literature (Hall and Lerner, 2010; Mohnen et al., 2017). As for the other parameters, the magnitudes of $\hat{\beta}_k$ and $\hat{\beta}_l$ are quite similar to the estimates of the elasticity of output to capital and labor in recent literature.[19]

[19]See for example De Loecker and Warzynski (2012), De Loecker et al. (2020), and Dobbelaere and Mairesse (2013). Note that the signs of $\hat{\beta}_k$ and $\hat{\beta}_l$ are negative because my model starts from the marginal cost (Equation (2)), where more capital and labor lower the firm's marginal cost. Revenue- or production-based models (e.g., De Loecker, 2011;

Again, take the example of the Dutch ICT firm that develops a dynamic pricing algorithm. If the firm builds the algorithm in-house, its productivity in the next period is on average about 5.0% higher than if it did no R&D. If it only consults university researchers, the gain is about 5.8%, and if it does both, about 5.4%. That is, the firm gains about the same from each mode. In particular, a firm that already does in-house R&D gains only about 0.4% more productivity by also contracting out, although the Wald test above cannot reject that this gain equals the 5.8% gain from contracting out alone. This result is important for the rest of the paper. Since doing both does not add a distinguishable productivity gain, any complementarity between the two modes has to come from the cost side, which I discuss in the next subsection.

The estimated parameters shown in Table 4 will be used to construct the (discretized) productivity of the dynamic model. The standard deviation from the (predicted) residuals, shown in the second to the last row of Table 4, will also be used to construct the transition probability.

5.2 Costs of Innovation and Complementarity

Finally, I discuss the three dynamic costs, which I estimate by maximum likelihood using the nested fixed-point algorithm of Rust (1987). Table 5 reports the estimates for the full sample (2000–2020), and for the periods before (2000–2008) and after (2012–2020) the expansion of the Innovation Box.[20] All costs are in millions of EUR.

Table 5: Costs of Innovation and Cost-Complementarity

Parameters	**Costs of R&D and Complementarity**		
	Full Period: 2000 - 2020	**Before Innovation Box: 2000 - 2008**	**After Innovation Box: 2012 - 2020**
$\kappa^{internal}$	0.127^{***}	0.569^{***}	-0.063
	(0.049)	(0.096)	(0.062)
$\kappa^{external}$	2.453^{***}	2.239^{***}	2.695^{***}
	(0.120)	(0.186)	(0.177)
$\kappa^{complement}$	2.295^{***}	2.153^{***}	2.502^{***}
	(0.132)	(0.217)	(0.188)
log-likelihood	-2836.652	-786.940	-1784.994

Notes.—The estimates are in millions of EUR (real). The gradients and Hessian are obtained by finite-difference approximation evaluated at the estimated parameters. Gradients are below $1e-4$. Tilt and radius of curvature for these parameters are below $1e-3$, computed by axial search at each parameter. Standard errors are obtained by computing the square root of the diagonal of the inverse Hessian.

Costs of internal and external R&D.—First, I find that external R&D is much more costly than

De Loecker and Warzynski, 2012; De Loecker et al., 2020) instead estimate positive output elasticities, since inputs directly increase revenue or output. Under cost minimization, the two approaches are dual (Varian, 1992), so the absolute magnitudes, $|\hat{\beta}_k| \approx 0.193$ and $|\hat{\beta}_l| \approx 0.650$, can be compared with the output elasticities of about 0.1 to 0.3 for capital and 0.5 to 0.7 for labor in those studies.

[20] I let the post-expansion period begin in 2012 to allow for a lag between the 2010 expansion and its use by firms. The reason is that firms may need time to learn about the new scheme, to apply for it, and to adjust their R&D. Consistent with such lags, Verhoeven and van Stel (2013) document a gradual increase in the take-up of the WBSO and the Patent Box among small firms from 2006 to 2010. See also Mohnen et al. (2017).

internal R&D. In the full sample, the cost of external R&D is 2.453 million EUR, while the cost of internal R&D is 0.127 million EUR, i.e., external R&D is about nineteen times as costly. Before the Innovation Box expansion, the two costs are 2.239 and 0.569 million EUR, so external R&D is about four times as costly as internal R&D. These estimates are close to the simple log ratios of the shares discussed in Section 4. The high cost of external R&D is among the main reasons why so few firms rely on external R&D alone: only about 3% of the observations in the sample choose external-only.

Cost-complementarity.—Second, I find that internal and external R&D are complementary on the cost side, i.e., $\kappa^{complement} > 0$, in all three periods. The estimated cost-complementarity is 2.295 million EUR in the full sample, 2.153 million EUR before the expansion, and 2.502 million EUR after. Take a firm that already performs internal R&D and is deciding whether to also contract part of its research out. On its own, external R&D would cost the firm 2.453 million EUR. Since the firm already does in-house R&D, 2.295 million EUR of that cost is saved, and the firm pays only about 0.16 million EUR extra. That is, the net cost of doing both is $\kappa^{internal} + \kappa^{external} - \kappa^{complement} = 0.285$ million EUR, against 0.127 million EUR for internal R&D alone. This is consistent with the 27.7% of the observations that choose both. As for the other two periods, the net cost of doing both is 0.655 million EUR before the expansion and 0.130 million EUR after, or 0.193 million EUR if I treat the post-expansion cost of internal R&D as zero.

Interpretation.—Why is doing both so much cheaper than the sum of the two costs? Recall from the previous subsection that the productivity gains are similar across modes, so the complementarity between the two modes has to come from the cost side. This is in line with Goyal et al. (2008), who show that complementarity can arise from cost advantages alone. In the hold-up interpretation of Section 3.3, the cost of external R&D is the value that a firm loses when it contracts out without its own R&D staff, and the cost-complementarity is the part of that loss that it recovers when it also does in-house R&D. The estimates then say that a firm with its own R&D staff recovers almost all of that loss: in the full sample, the cost-complementarity offsets $2.295/2.453 \approx 94\%$ of the cost of external R&D. Returning to the example, the firm that builds its pricing algorithm in-house can consult university researchers at little extra cost, while a firm without its own R&D staff faces the full cost of contracting out. This result is related to the evidence that in-house R&D raises the return to external knowledge (Cassiman and Veugelers, 2006; Cohen and Levinthal, 1990; Schmiedeberg, 2008; Lokshin et al., 2008).

Share of the external cost that is offset.—Table 6 reports this share, $r = \kappa^{complement}/\kappa^{external}$. I find the following: 0.962 before the Innovation Box expansion, 0.936 in the full sample, and 0.928 after. Since r is a ratio of two estimated costs, it does not require a value of the surplus scale S of Section 3.3. In the hold-up model, this ratio also gives a lower bound on the firm's share of the collaboration when it does both. Since $(\tau')^2 = r + (1-r)\tau^2 \geq r$, this share is at least $\sqrt{r}$, which is 0.967 in the full sample, 0.981 before the expansion, and 0.964 after. That is, if the estimated costs come from hold-up, a firm that also does in-house R&D keeps almost the entire value of the

collaboration. Appendix J.4 shows how τ varies with S.

Table 6: How much of the external cost is offset by also doing in-house R&D

Sample window	$\kappa^{external}$	$\kappa^{complement}$	r	Lowest τ'
Before Innovation Box, 2000–2008	2.239	2.153	0.962	0.981
Full period, 2000–2020	2.453	2.295	0.936	0.967
After Innovation Box, 2012–2020	2.695	2.502	0.928	0.964

Notes. Costs are the point estimates from Table 5, in million EUR. $r = \kappa^{complement}/\kappa^{external}$ is the share of the extra cost of contracting out that complementarity offsets. The last column is $\sqrt{r}$, the lowest bargaining share when the firm does both that is consistent with that r (Section 3.3).

Changes after the Innovation Box expansion.—After the expansion, the cost of internal R&D is -0.063 million EUR and statistically indistinguishable from zero.[21] I treat this estimate as about zero when describing policy. One reading of this fall is the WBSO. By lowering the wage taxes on the firm's own R&D staff, the government may have indirectly paid for a large part of the cost of in-house R&D. Meanwhile, the cost of external R&D rises to 2.695 million EUR after the expansion. I argue that this increase is related to the Nexus Approach of the Innovation Box, under which only a limited part of the firm's outsourced R&D still qualifies for the scheme (see Section 2). The cost-complementarity rises with it, from 2.153 to 2.502 million EUR, so the share r in Table 6 stays high. That is, after the expansion, contracting out without in-house R&D becomes more expensive, while contracting out together with in-house R&D remains cheap.

6 Simulation Analysis

This section uses the estimated model to simulate R&D subsidies that try to mimic some elements of the Dutch Tax Incentives for Innovation scheme. In particular, I approach the scheme as a reduction in the costs of R&D, $\kappa^{internal}$ and $\kappa^{external}$. I investigate two things: (i) what happens to the share of firms in each R&D mode following such a subsidy, and (ii) what happens to the firm's value. I also explore what happens to these subsidies when the cost-complementarity between internal and external R&D is shut off. Recall from Section 2 that the Dutch scheme tilts toward in-house R&D, since under the OECD Nexus Approach only a limited part of outsourced R&D still qualifies for the Innovation Box (OECD, 2015).

To compute the counterfactuals, I re-solve the model under the new costs and iterate to a new steady-state distribution of states and choices. I then report the change in the share of firms in each R&D mode, and the change in firm value, $\Delta V^c = (V^c - V)/V$, where V is the ex-ante expected value from Equation (6), i.e., McFadden's social surplus. Since I assume monopolistic

[21] The statistically insignificant estimate partly results from less variation in R&D choices after the expansion. As shown in Fact 2, the share of firms that choose no R&D falls after 2010, which leaves fewer firms whose choice between no R&D and internal R&D pins down the cost of internal R&D. This weakens identification and inflates the standard error.

competition, this change in firm value can be thought of as the change in producer surplus in partial equilibrium.

I consider three cases. The first case is a uniform 5% reduction in the costs of both internal and external R&D. This case reflects a subsidy that does not favor any R&D mode. The second case is a 20% reduction in the cost of internal R&D, while keeping the cost of external R&D the same. This case tries to mimic the tilt of the Dutch scheme toward in-house R&D. Finally, the third case is a 5% reduction in the cost of external R&D only. The Dutch scheme does not subsidize contracted-out R&D, so this third case serves as a comparison. Note that, because the cost of external R&D is much larger than the cost of internal R&D, a 5% cut in the cost of external R&D is larger in euros than a 20% cut in the cost of internal R&D: about 0.12 versus 0.03 million EUR under the full-sample estimates. For each case, I also re-solve the model with the cost-complementarity set to zero, $\kappa^{complement} = 0$. In the hold-up interpretation, this means that in-house R&D no longer raises the share of the value that the firm keeps from the collaboration, i.e., $\tau' = \tau$.

6.1 Change in the Share of R&D activities

Table 7 reports the change in the steady-state share of each R&D mode relative to the actual shares in the sample. Throughout the sample period, 36.8% of the observations perform no R&D at all, 32.4% perform only internal R&D, 3.2% perform only external R&D, and 27.7% perform both. I find that the uniform 5% subsidy (Case 1) yields the largest increase in the share of R&D-active firms. The share of firms that perform no R&D falls by 1.6 percentage points, and the share performing only internal R&D falls by 1.2 percentage points. At the same time, the share performing only external R&D rises by 0.2 percentage points, and the share performing both rises by 2.4 percentage points. Overall, the result suggests that some firms that initially choose not to do R&D now perform one or both types of R&D, and that some firms that only perform internal R&D now also contract part of their research out.

Table 7: Change in the share of R&D activities

Change in the Share of R&D Activities	**Actual Share**	$\Delta^c = share^c - share$					
		5% uniform		**20% internal**		**5% external**	
		Base	***No Comp.***	***Base***	***No Comp.***	***Base***	***No Comp.***
No R&D	0.368	-0.016	0.115	-0.006	0.116	-0.015	0.116
Internal-only	0.324	-0.012	0.104	0.003	0.113	-0.012	0.103
External-only	0.032	0.002	0.015	-0.001	0.01	0.002	0.015
Both	0.277	0.024	-0.235	0.003	-0.239	0.024	-0.235

Notes.—Base: changes only to the costs. No Comp: Set the cost-complementarity parameter to zero, i.e., $\kappa^{complement} = 0$. Negative sign indicates "decreasing".

The internal-only 20% subsidy (Case 2) has small effects. The share of firms that perform no R&D falls by only 0.6 percentage points, the shares performing only internal R&D and both each rise by 0.3 percentage points, and the share performing only external R&D falls slightly, by 0.1

percentage points. The main reason is that the cost of internal R&D is already low, 0.127 million EUR, so a 20% cut makes little difference to the firm's choice. In the hold-up interpretation, this subsidy makes in-house R&D cheaper, but it leaves the extra cost of contracting out without in-house R&D unchanged. As for the external-only 5% subsidy (Case 3), it ranks second, after the uniform subsidy. The share of firms that perform no R&D falls by 1.5 percentage points, the share performing only internal R&D falls by 1.2 percentage points, the share performing only external R&D rises by 0.2 percentage points, and the share performing both rises by 2.4 percentage points. That is, cutting the cost of external R&D mostly moves firms into doing both.

When cost-complementarity is shut off, the picture changes completely. Take the uniform subsidy. Instead of falling, the share of firms that perform no R&D rises by 11.5 percentage points, and the share performing only internal R&D rises by 10.4 percentage points. Meanwhile, the share performing both falls by 23.5 percentage points. The same pattern appears under the other two subsidies: the share performing both falls by 23.9 percentage points under the internal-only subsidy and by 23.5 percentage points under the external-only subsidy. That is, without cost-complementarity, most firms that do both would stop contracting out, and many would stop doing R&D at all. The reason is that doing both then costs the sum of the cost of internal R&D and the full cost of external R&D, which makes it much less attractive.

6.2 Change in Firm's Valuation

Table 8 shows the average change in firm value, ΔV^c. I find that the uniform subsidy yields the largest gain, 2.74%. The external-only subsidy follows closely, at 2.49%. Meanwhile, the internal-only subsidy raises firm value by only 0.97%. The ranking follows the same logic as for the shares. A cut in the cost of external R&D lowers the largest cost that firms face, while a cut in the cost of internal R&D lowers a cost that is already small.

Table 8: Change in Value

	$\Delta V^c = (V^c - V)/V$					
	5% uniform		**20% internal**		**5% external**	
Change in Value	***Base***	***No Comp.***	***Base***	***No Comp.***	***Base***	***No Comp.***
	2.735%	1.022%	0.968%	0.923%	2.49%	0.792%

Notes.—Base: changes only to the costs. No Comp: Set the cost-complementarity to zero. Negative sign indicates "decreasing".

Without cost-complementarity, the gains in firm value fall to 1.02% under the uniform subsidy, 0.92% under the internal-only subsidy, and 0.79% under the external-only subsidy. That is, the external-only subsidy loses most of its effect, from 2.49% to 0.79%, while the internal-only subsidy barely changes, from 0.97% to 0.92%. The reason is that, without cost-complementarity, cheaper external R&D no longer makes doing both attractive for firms that already do in-house R&D. Overall, to achieve the same gain in firm value without cost-complementarity, the government would have to spend more on the subsidy.

To conclude, a uniform subsidy raises R&D participation and firm value by more than a subsidy that only lowers the cost of in-house R&D. This result relates to the literature on the design of R&D policies (Galaasen and Irarrazabal, 2021; Akcigit et al., 2021, 2022). In particular, Galaasen and Irarrazabal (2021) find that a uniform subsidy to Norwegian firms raises investment, growth, and welfare by more than a size-dependent subsidy. In the hold-up interpretation, a subsidy aimed only at in-house R&D does little, since in-house R&D is already cheap, and it leaves in place the high cost that firms face when they contract out without their own R&D staff.

Finally, the hold-up interpretation also points to another type of policy, one that strengthens the firm's position in its external R&D contracts. Given a value of S, such a policy would move the cost of external R&D and the cost-complementarity together (Appendix J.4). Evaluating it would require re-solving the model for each chosen value of S, which I leave for future work.

7 Conclusion

This paper studies the costs and benefits of in-house and contracted-out R&D in the Dutch ICT industry from 2000 to 2020. I first document that most R&D-active firms conduct R&D in-house, either alone or together with external partners, and that very few firms rely on contracted-out R&D alone. To rationalize these facts, I build and estimate a dynamic model in which each R&D mode has its own cost and shifts the firm's future productivity, and in which doing both can be cheaper than the sum of the two costs.

I find three main results. First, the productivity gains are similar across the R&D modes. Second, contracted-out R&D is much more costly than in-house R&D, which is among the main reasons why so few firms contract out alone. Third, doing both is much cheaper than the sum of the two costs: for a firm that already does in-house R&D, most of the extra cost of contracting out disappears. I interpret this pattern through a hold-up problem on the external contract, in which a firm with its own R&D staff keeps most of the value of the collaboration. After the Innovation Box expansion, the cost of in-house R&D falls to about zero, while the cost of contracted-out R&D rises. This is consistent with the WBSO lowering the cost of the firm's own R&D staff, and with the Nexus Approach limiting the support for outsourced R&D.

In the simulations, a uniform cut in R&D costs raises R&D participation and firm value by more than a cut that only lowers the cost of in-house R&D, since the cost of in-house R&D is already low. When cost-complementarity is shut off, the share of firms that do both falls sharply, and the gains in firm value from all three subsidies shrink.

The paper has four limitations. First, I model Dutch ICT firms as monopolistic competitors, so one firm's R&D choice does not change its rivals' profits. Strategic interactions, such as competing for the same partner, waiting to see a rival's product, or contracting with a partner to keep it away from a rival, are then absorbed into the estimated costs. For example, if firms compete for the same partners, part of the estimated cost of contracted-out R&D reflects that competition, in

addition to hold-up.[22] Second, the hold-up problem is one interpretation of the estimated costs. Haggling over incomplete contracts would produce the same pattern of costs, and since the data record only the R&D modes and not the contracts themselves, I cannot distinguish between the two explanations. Third, the simulations are in partial equilibrium, and I only report the change in firm value, i.e., producer surplus. Fourth, the estimates apply to Dutch ICT firms, and the CIS only records whether a firm uses each R&D mode, and it does not record how much the firm spends on each mode. The costs are therefore inferred from the firms' choices, and they are not observed expenditures. As for future research, I would like to complement this ex-ante evaluation with an ex-post evaluation of the Dutch Tax Incentives for Innovation scheme. I would also like to study whether the scheme creates incentives for firms to relabel other spending as innovation-related.

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A Probit: Persistence of R&D modes

Table 9: Probit: Persistence of R&D modes

Probit	***Dep. var***			
	No R&D (t)	**Internal-only (t)**	**External-only (t)**	**Both (t)**
No R&D (t-1)	1.171***			
	(0.0895)			
Internal-only (t-1)		0.7635***		
		(0.0950)		
External-only (t-1)			1.198***	
			(0.2105)	
Both (t-1)				1.017***
				(0.0821)
Year FE	Yes	Yes	Yes	Yes
Sub-Industry FE	Yes	Yes	Yes	Yes

B Details on the Reduced-form Testing of Complementarity between Internal and External R&D

I now provide a simple, reduced-form test for complementarity between $InternalR\&D$ and $ExternalR\&D$. To do so, I follow similar approaches used by Arora (1996), Athey and Stern (1998), and Cassiman and Veugelers (2006). First, let us define Π as a firm's profit function that is assumed to be supermodular. The arguments of this profit function are $InternalR\&D$ and $ExternalR\&D$. Then, as shown by Milgrom and Roberts (1990, 1995), $InternalR\&D$ and $ExternalR\&D$ are complementary only if

$$\Pi(\text{both R\&D}) - \Pi(\text{external}) \geq \Pi(\text{internal}) - \Pi(\text{no R\&D}) \tag{19}$$

The main intuition of Equation (19) is that adding an activity while the other activity is already being performed has a higher incremental effect on profits than adding the activity in isolation. Here, performing external R&D on top of internal R&D can only be deemed more profitable (compared to profits from performing *only* internal R&D) if the above condition holds.

In the empirical implementation, I run the following specification

$$\begin{aligned} y_{ijt} = {} & \theta_{\text{both}} BothR\&D_{ijt} + \theta_{\text{internal}} InternalR\&D_{ijt} \\ & + \theta_{\text{external}} ExternalR\&D_{ijt} + \theta_{\text{no}} NoR\&D_{ijt} + \zeta_t + \zeta_j + \varepsilon_{ijt} \end{aligned} \tag{20}$$

where the dependent variable y_{ijt} is either value-added or the share of new products produced by each firm in a given year. ζ_t indicates year fixed effects. Since the share of new products in revenue ranges from 0 to 100, a standard OLS will yield biased estimates. Given this concern, and the ability to exploit the panel structure of the model, I estimate Equation (20) using a Panel (Random Effect) Tobit model.

Testing for the complementarity boils down to checking if

$$\theta_{\text{both}} - \theta_{\text{internal}} \geq \theta_{\text{external}} - \theta_{\text{no}} \tag{21}$$

Complementarity is $\Delta \equiv (\theta_{\text{both}} - \theta_{\text{internal}}) - (\theta_{\text{external}} - \theta_{\text{no}}) \geq 0$. The last row of Table 3 is a Wald test of H_0: $\Delta = 0$, using the χ^2 statistic of that linear restriction and the sign of $\widehat{\Delta}$. The point estimates are $\widehat{\Delta} = -0.200$ for log value added, $\widehat{\Delta} = 4.74$ for the share of new products under OLS, and $\widehat{\Delta} = -0.57$ under the random-effects Tobit. None of the three tests rejects $\Delta = 0$. The same difference on the productivity parameters in Table 4 is $\widehat{\Delta}^{\omega} = (\alpha_8 - \alpha_6) - (\alpha_7 - \alpha_5) = -0.042$, with $p = 0.54$. Complementarity in the dynamic model is a cost object, identified from choice frequencies, not from these payoff-side differences.

C Details on the Derivation of the Static Part

Proof of Result 1: Revenue Equation.. First, recall the demand function is the following:

$$\begin{aligned} Q_{it} &= Q_t \left(\frac{P_{it}}{P_t} \right)^{\eta} \\ &= \frac{Q_t}{P_t^{\eta}} P_{it}^{\eta} \\ &= \Phi_t P_{it}^{\eta} \end{aligned} \tag{22}$$

Each firm maximizes its profit,

$$\pi_{it} = P_{it} Q_{it} - C_{it} Q_{it} \tag{23}$$

We can substitute Q_{it} using the expression we have in the demand function, we have the following.

$$\pi_{it} = \Phi_t P_{it}^{1+\eta} - C_{it} \Phi_t P_{it}^{\eta} \tag{24}$$

Take the first-order condition with respect to P_{it}, we have the following.

$$\frac{\partial \pi_{it}}{\partial P_{it}} = (1+\eta) \Phi_t P_{it}^{\eta} - \eta C_{it} \Phi_t P_{it}^{\eta - 1} = 0 \tag{25}$$

Simple algebra of the above first-order condition provides us with the following relation between price P_{it} and marginal cost C_{it}.

$$P_{it} = \frac{\eta}{\eta + 1} C_{it} \tag{26}$$

Now, revenue can be expressed as $R_{it} = P_{it} Q_{it}$. We can further re-write the revenue equation, using both the simplified expression of demand function and the relationship between price and

marginal cost, as follows.

$$\begin{aligned} R_{it} &= P_{it}Q_{it} \\ &= \Phi_t P_{it}^{1+\eta} \\ &= \Phi_t \left(\frac{\eta}{\eta+1} C_{it} \right)^{1+\eta} \end{aligned} \tag{27}$$

Take the log of the above expression and recall the functional form of log of marginal cost, $ln(C_{it}) = c_{it}$, we have the following.

$$\begin{aligned} r_{it} &= ln(R_{it}) \\ &= (1+\eta) ln \left(\frac{\eta}{\eta+1} \right) + ln(\Phi_t) \\ &+ (1+\eta) \Big[\beta_0 + \beta_k k_{it} + \beta_l l_{it} + \beta_w ln(w_t) - \omega_{it} \Big] \end{aligned} \tag{28}$$

We now have the final expression of Result 1. ■

Proof of Result 2: Static Profit Equation.. First, we define profit as follows.

$$\pi_{it} = P_{it}Q_{it} - C_{it}Q_{it} \tag{29}$$

using the relation between price and marginal cost, we can rewrite the above expression as follows.

$$\begin{aligned} \pi_{it} &= P_{it}Q_{it} - \frac{\eta+1}{\eta} P_{it}Q_{it} \\ &= R_{it} - \frac{\eta+1}{\eta} R_{it} \\ &= -\frac{1}{\eta} R_{it} \end{aligned} \tag{30}$$

Then, using the expression derived in Result 1, we have the following.

$$\pi_{it} = -\frac{1}{\eta} exp(r_{it}(k_{it}, l_{it}, \omega_{it}, w_t, \Phi_t; \eta, \beta_0, \beta_k, \beta_l, \beta_w)) \tag{31}$$

As we will see later, r_{it} can be reduced further to only depend only on k_{it}, l_{it}, ω_{it}, and parameters $\eta, \beta_0, \beta_k, \beta_l$. In other words, w_t and Φ_t will be absorbed as time-specific effects. Therefore, we can now have the final expression of π_{it} as follows.

$$\pi_{it} = -\frac{1}{\eta} exp(r_{it}(k_{it}, l_{it}, \omega_{it}; \eta, \beta_0, \beta_k, \beta_l)) \tag{32}$$

We now have the final expression of Result 2.

■

D Details on the Productivity Estimation

First, the expression in Equation (3) is appended with an IID error term u_{it}. That is,

$$\begin{aligned} r_{it} &= (\eta+1)ln\left(\frac{\eta}{\eta+1}\right) + ln(\Phi_t) \\ &+ (\eta+1)\Big[\beta_0 + \beta_k k_{it} + \beta_l l_{it} + \beta_w ln(w_t)\Big] \underbrace{-(\eta+1)\omega_{it} + u_{it}}_{\text{Composite Error Term}} \end{aligned} \tag{33}$$

where $\Phi_t := \frac{Q_t}{P_t^y}$.

I borrow insights from Olley and Pakes (1996) to re-write the unobserved productivity, ω_{it}, in terms of some observables which are correlated with it. In particular, as in Levinsohn and Petrin (2003); Ackerberg et al. (2015), and Gandhi et al. (2020), I use the firm's intermediate input, m_{it}, and the firm's energy input, n_{it}, as firm i's choice of this variable input depends on the level of productivity. The essence of Olley and Pakes (1996) is that this observable, i.e., intermediate input expenditures of the firm, contains information on its productivity level[23]. Then, I can write the level of productivity, conditional on the capital stock, as a function of the variable input levels, $\omega_{it}(k_{it}, l_{it}, m_{it}, n_{it})$. This method allows me to use the intermediate inputs of the firm to control for the productivity in Equation (33). I then combine the demand elasticity terms into a parameter γ_0, and the time-varying aggregate demand shock and market-level factor prices into a set of time and industry dummies ($D_t^{(1)}$ and $D_j^{(2)}$), I can re-write Equation (33) as follows

$$\begin{aligned} r_{it} &= \gamma_0 + \sum_{t=1}^{T} \gamma_t^{(1)} D_t^{(1)} + \sum_{j=1}^{J} \gamma_j^{(2)} D_j^{(2)} + (\eta+1)\Big[\beta_k k_{it} + \beta_l l_{it} - \omega_{it}\Big] + u_{it} \\ &= \gamma_0 + \sum_{t=1}^{T} \gamma_t^{(1)} D_t^{(1)} + \sum_{j=1}^{J} \gamma_j^{(2)} D_j^{(2)} + h(k_{it}, l_{it}, m_{it}, n_{it}) + \nu_{it} \end{aligned} \tag{34}$$

The function $h(\cdot)$ represents the combined effect of capital and productivity. In particular, I approximate this function with third-order polynomials. I estimate Equation (34) with a simple OLS. From here, I can get a fitted value of the $h(\cdot)$ where I denote it as $\hat{\phi}_{it}$. This $\hat{\phi}_{it}$ is an estimate of $(\eta+1)\Big[\beta_k k_{it} + \beta_l l_{it} - \omega_{it}\Big]$.

As discussed in the previous section, I assume that ω_{it} follows a controlled Markov process. I assume the distribution of ω_{it} can be written as $G_\omega(\cdot)$. For the empirical implementation of my model, I model the evolution of ω_{it} as follows

$$\begin{aligned} \omega_{it} &= g_\omega(\omega_{it-1}, \text{no}_{it-1}, \text{int}_{it-1}, \text{ext}_{it-1}, \text{both}_{it-1}) + \xi_{it} \\ &= \alpha_0 + \alpha_1 \omega_{it-1} + \alpha_2 (\omega_{it-1})^2 + \alpha_3 (\omega_{it-1})^3 \\ &+ \alpha_4 \text{no}_{it-1} + \alpha_5 \text{int}_{it-1} + \alpha_6 \text{ext}_{it-1} + \alpha_7 \text{both}_{it-1} + \xi_{it} \end{aligned} \tag{35}$$

[23]There are two crucial assumptions for this insight to work. The first one is the scalar unobservability assumption. The intuition behind this assumption is that in choosing the input, firms only require one unobservable (in our case ω_{it}). The second assumption is the strictly monotone assumption. This assumption dictates that the intermediate input must be strictly monotone in ω. Levinsohn and Petrin (2003) show that the assumption is not too restrictive in the case of manufacturing plants in Chile. These assumptions allow us to invert the intermediate input, $m_{it} = \mathbb{M}_t(k_{it}, l_{it}, \omega_{it})$.

Here, I assume that ξ_{it} is IID across time and firms and is drawn from a normal distribution with zero mean and variance σ^2_{ξ}.

I can now substitute the productivity ω_{it} using $\hat{\phi}_{it}$. That is, substituting $\omega_{it} = -(1/(\eta+1))\hat{\phi}_{it} + \beta_k k_{it} + \beta_l l_{it}$ to the Equation (35). After some algebra, I obtain the following estimating equation

$$\begin{aligned}\hat{\phi}_{it} = {} & \beta_k^* k_{it} + \beta_l^* l_{it} + \alpha_1 \left(\hat{\phi}_{it-1} - \beta_k^* k_{it-1} - \beta_l^* l_{it-1}\right) \\ & - \alpha_2^* \left(\hat{\phi}_{it-1} - \beta_k^* k_{it-1} - \beta_l^* l_{it-1}\right)^2 \\ & + \alpha_3^* \left(\hat{\phi}_{it-1} - \beta_k^* k_{it-1} - \beta_l^* l_{it-1}\right)^3 \\ & - \alpha_4^* \text{no}_{it-1} - \alpha_5^* \text{internal}_{it-1} - \alpha_6^* \text{external}_{it-1} - \alpha_7^* \text{both}_{it-1} - \xi_{it}^*\end{aligned} \tag{36}$$

where the star, $(*)$, represents that the coefficients α and β are multiplied by $(\eta+1)$[24]. The above expression will be estimated with nonlinear least squares. From the estimated parameters, I can recover an estimate of productivity for each observation as follows. The factor $-1/(\eta+1)$ is also how the mode parameters in Table 4 are mapped into a productivity difference in Section 5.

$$\hat{\omega}_{it} = -(1/(\hat{\eta}+1))\hat{\phi}_{it} + \hat{\beta}_k k_{it} + \hat{\beta}_l l_{it} \tag{37}$$

Source of identification.—This approach is essentially a semiparametric one. As noted by Doraszelski and Jaumandreu (2013), the main ingredient for identification is that variables in the parametric part of the model are not perfectly predictable by the variables in the non-parametric part (in a simple regression sense). Apart from a similar identification as in Olley and Pakes (1996) and Doraszelski and Jaumandreu (2013), the estimating equation in Equation (36) makes use of information on the demand elasticity. This demand elasticity is identified, and the source of its identification has been discussed in the previous subsection.

E Alternative Demand System

I follow a standard discrete choice model of consumer choice a la Berry (1994) and Berry et al. (1995). The indirect utility function can be described as follows

$$V_{ijt} = \eta p_{jt} + \xi_{ijt} + \varepsilon_{jt} \tag{38}$$

where p_{jt} is the logarithmic price, ξ_{ijt} is the idiosyncratic preference shock which follows a Type-1 Extreme Value distribution, and ε_{jt} is an unobserved demand shock.

Using the Berry inversion as in Berry (1994) for the above expression, I can get a well-known expression for the market share of good j relative to the outside option as follows

$$ln(ms_{jt}) - ln(ms_{ot}) = \eta p_{jt} + \xi_{ijt} + \varepsilon_{jt} \tag{39}$$

Consider only the case of single-product firm ($i = j$). From the log of market share, I can rearrange the above expression using that $ln(ms_{it}) = ln(q_{it}) - ln(Q_t)$, as in the demand expressed in Equation (1). Then, I obtain the expression for log price $ln(p_{jt})$. Note that $ln(r_{it}) = ln(p_{it}q_{it})$.

[24] With some exceptions for $\alpha_2^* = \alpha_2(\eta+1)^{-1}$ and $\alpha_3^* = \alpha_3(\eta+1)^{-2}$

Rearranging this expression with information on the expression of $ln(p_{it})$ and Equation (1), this yields the following expression for log revenue

$$ln(r_{it}) = \kappa_0 + \kappa_k ln(k_{it}) + \omega_{it}^* + u_{it} \tag{40}$$

where $\kappa_0 = \frac{1}{|\eta|} ln(ms_{ot})$, $\kappa_k = (\eta + 1)\beta_k$, and $\omega_{it}^* = -(\eta + 1)\omega_{it}$. The above expression is similar to the estimating equation denoted in Equation (34) in the absence of the year fixed effects. As noted by De Loecker (2011), the total output enters in exactly the same way, leading to identification of η in the production function framework. Anderson et al. (1992) note that in the usual logit demand structure, the estimated parameter η is used to compute own and cross-price elasticities. However, in this setup with log prices in the indirect utility function, they are identical.

F Derivation of the McFadden's Social Surplus

Under the Type 1 Extreme Value distribution assumption for ε_{it}^a, there is a closed-form expression for the expected value of the **max** over options. This is a standard result from McFadden (1978) for static logit models, extended to dynamic settings by Rust (1987),

Let us denote the deterministic part of each action's payoff, excluding the shock, as follows:

$$\begin{aligned}
\bar{V}_{it}^{no} &= \beta EV^{no} \\
\bar{V}_{it}^{internal} &= -\kappa^{internal} + \beta EV^{internal} \\
\bar{V}_{it}^{external} &= -\kappa^{external} + \beta EV^{external} \\
\bar{V}_{it}^{both} &= -\kappa^{internal} - \kappa^{external} + \kappa^{complement} + \beta EV^{both}
\end{aligned}$$

The realized value is $V(s_{it}) = \pi_{it}(s_{it}) + \max_a\{\bar{V}_{it}^a + \varepsilon_{it}^a\}$. Then, the ex-ante expectation is:

$$EV(s_{it}) = \pi_{it}(s_{it}) + E_\varepsilon \left[\max_a \left(\bar{V}_{it}^a + \varepsilon_{it}^a\right)\right]$$

where $\gamma^{Euler} \approx 0.57721$ is Euler's constant from the Type 1 Extreme Value distribution's properties.

For an IID Type 1 EV shock with scale 1, the expected max has a closed-form solution:

$$E_\varepsilon \left[\max_a \left(v_a + \epsilon_a\right)\right] = ln\left(\sum_a exp(v_a)\right) + \gamma^{Euler}$$

Then, we have that the ex-ante expectation is the following

$$\begin{aligned}
EV(s_{it}) = \pi_{it}(s_{it}) &+ \gamma^{\text{Euler}} \\
&+ ln\left[exp(\overline{V}_{it}^{no}) + exp(\overline{V}_{it}^{internal}) + exp(\overline{V}_{it}^{external}) + exp(\overline{V}_{it}^{both})\right]
\end{aligned}$$

G Derivation of the Expected Future Value

The value function $V(s_{it})$ represents the maximum expected discounted lifetime profits starting from state s_{it}, following optimal behavior. After choosing action a, the firm receives current profit $\pi(s_{it})$ minus costs, plus the discounted expected value from $t+1$ onward.

The expected future value conditional on action a is defined as the expectation of $V(s_{it+1})$ over the distribution of next states:

$$EV^a(s_{it}) = \mathbb{E}[V(s_{it+1} \mid s_{it}, a]$$

This is the standard definition in the typical dynamic discrete choice model (see Rust (1987), Abbring and Daljord (2020)). It captures how the choice of a influences the evolution of the state and thus future values.

Assuming that the state space is continuous, we can express the expectation using the transition density $p(s_{it+1}|s_{it}, a)$:

$$EV^a(s_{it}) = \int_s V(s_{it+1}) p(s_{it+1} \mid s_{it}, a) ds$$

Then, in our case,

$$\begin{aligned} EV^a(s_{it}) &= \int_{\omega'} V(s_{it+1}) dG(\omega_{it+1} \mid s_{it}, a_{it}) \\ &, \qquad \forall a_{it} \in \{\text{no R\&D, internal, external, both}\} \end{aligned}$$

Since, I binned the state, we need to model the state space as a discrete one. In that case, the expected future value has the following expression.

$$\begin{aligned} EV^a(s_{it}) &= \int_s V(s_{it+1}) p(s_{it+1} \mid s_{it}, a) ds \\ &\approx \sum_s V(s_{it+1}) p(s_{it+1} \mid s_{it}, a) \end{aligned}$$

Similarly, if the continuous state space (e.g., productivity ω) is discretized into a finite grid for computational tractability, the expected future value can be approximated as a discrete sum over possible next states. This is computationally efficient, as it can be represented as a matrix multiplication $EV^a = P^a \cdot V$, where P^a is the action-specific transition matrix and V is the vector of value functions. It preserves action-dependence: Different actions (e.g., internal R&D vs. no R&D) lead to different transition probabilities P^a, reflecting how R&D choices affect state evolution (e.g., higher probability of jumping to a high-innovation state with both types of R&D).

$$\begin{aligned} EV^a(s_{it}) &= \int_{\omega'} V(s_{it+1})\, dG(\omega' \mid s_{it}, a_{it}) \\ &\approx \sum_{\omega'} V(s_{it+1})\, P(\omega' \mid s_{it}, a_{it}) \\ &, \qquad \forall a_{it} \in \{\text{no R\&D, internal, external, both}\} \end{aligned}$$

This is computationally efficient since it can be represented simply as a matrix multiplication. It preserves action-dependence: Different actions (e.g., "internal R&D" vs. "no R&D") lead to different transition probabilities P^a, reflecting how R&D choices affect state evolution (e.g., higher probability of jumping to a "high innovation" state with "both" R&D).

H Derivation of the Probability

As I have outlined in the main text, suppose $\varepsilon_{it}(a_{it})$ is IID that follows a Type 1 Extreme Value distribution. The density of each private shock is $f(\varepsilon_{it}) = e^{-\varepsilon_{it}} e^{-e^{-\varepsilon_{it}}}$. The cumulative distribution is $F(\varepsilon_{it}) = e^{-e^{-\varepsilon_{it}}}$.

Suppose firm i chooses action *both*. In choosing this particular alternative, it must be that the value of doing *both* is higher than other alternatives than itself. We can express it as follows

$$\begin{aligned} Pr(a_{it} = both) &= Prob\left(\overline{V}_{it}^{both} + \varepsilon_{it}^{both} > \overline{V}_{it}^{j} + \varepsilon_{it}^{j}, \forall j \neq both\right) \\ &= Prob\left(\varepsilon_{it}^{j} < \varepsilon_{it}^{both} + \overline{V}_{it}^{both} - \overline{V}_{it}^{j}, \forall j \neq both\right) \end{aligned}$$

where $\overline{V}_{it}^{both}$ and $\overline{V}_{it}^{j}, \forall j \neq both$ are the deterministic parts of the conditional (alternative-specific) values.

Since ε's are independent, we can express the conditional probability as follows

$$Pr(a_{it} = both | \varepsilon_{it}^{both}) = \prod_{j \neq both} e^{-e^{-\left(\varepsilon_{it}^{both} + \overline{V}_{it}^{both} - \overline{V}_{it}^{j}\right)}}$$

Then,

$$\begin{aligned} Pr(a_{it} = both) &= \int \left(Pr(a_{it} = both | \varepsilon_{it}^{both})\right) f(\varepsilon_{it}^{both}) d\varepsilon_{it}^{both} \\ &= \int \left(\prod_{j \neq both} e^{-e^{-\left(\varepsilon_{it}^{both} + \overline{V}_{it}^{both} - \overline{V}_{it}^{j}\right)}} \right) e^{-\varepsilon_{it}^{both}} e^{-e^{-\varepsilon_{it}^{both}}} d\varepsilon_{it}^{both} \\ &= \int \left(\prod_{j} e^{-e^{-\left(\varepsilon_{it}^{both} + \overline{V}_{it}^{both} - \overline{V}_{it}^{j}\right)}} \right) e^{-\varepsilon_{it}^{both}} d\varepsilon_{it}^{both} \\ &= \int exp\left(-\sum_{j} e^{-\left(\varepsilon_{it}^{both} + \overline{V}_{it}^{both} - \overline{V}_{it}^{j}\right)} \right) e^{-\varepsilon_{it}^{both}} d\varepsilon_{it}^{both} \\ &= \int exp\left(-e^{-\varepsilon_{it}^{both}} \sum_{j} e^{-\left(\overline{V}_{it}^{both} - \overline{V}_{it}^{j}\right)} \right) e^{-\varepsilon_{it}^{both}} d\varepsilon_{it}^{both} \end{aligned}$$

Now suppose $x = exp(-\varepsilon_{it}^{both})$. We also know that $-exp(-\varepsilon_{it}^{both}) d\varepsilon_{it}^{both} = dx$. As ε_{it}^{both} approaches infinity, x approaches zero. Similarly, as ε_{it}^{both} approaches negative infinity, x becomes infinitely

large. Replacing the above expression with the new term, we have the following

$$\begin{aligned} Pr(a_{it} = both) &= \int_{\infty}^{0} exp\left(-x \sum_j e^{-\left(\overline{V}_{it}^{both} - \overline{V}_{it}^{j}\right)}\right)(-dx) \\ &= \int_{0}^{\infty} exp\left(-x \sum_j e^{-\left(\overline{V}_{it}^{both} - \overline{V}_{it}^{j}\right)}\right) dx \end{aligned}$$

Evaluating the integral in the last line, we arrive at the following expression

$$\begin{aligned} Pr(a_{it} = both) &= \frac{exp\left(-x \sum_j e^{-\left(\overline{V}_{it}^{both} - \overline{V}_{it}^{j}\right)}\right)}{-\sum_j e^{-\left(\overline{V}_{it}^{both} - \overline{V}_{it}^{j}\right)}}\Bigg|_0^{\infty} \\ &= \frac{1}{\sum_j e^{-\left(\overline{V}_{it}^{both} - \overline{V}_{it}^{j}\right)}} \\ &= \frac{e^{\overline{V}_{it}^{both}}}{\sum_j e^{\overline{V}_{it}^{j}}} \end{aligned}$$

Similarly, we can derive the optimal choice probability for other actions as well.

I Identification of the three R&D costs

I estimate the three costs by the nested fixed-point in Section 4. The inversions below follow the conditional-choice-probability map of Hotz and Miller (1993).

The current-period cost of no R&D is zero. The Type-1 extreme-value shocks have scale one. The discount factor is $\beta = 0.95$ and is not estimated.

Equation (8) implies, for each state,

$$\begin{aligned} \log\Big(\frac{P^{internal}}{P^{none}}\Big) &= -\kappa^{internal} + \beta\big(EV^{internal} - EV^{none}\big), \\ \log\Big(\frac{P^{external}}{P^{none}}\Big) &= -\kappa^{external} + \beta\big(EV^{external} - EV^{none}\big), \\ \log\Big(\frac{P^{both}}{P^{none}}\Big) &= -\kappa^{internal} - \kappa^{external} + \kappa^{complement} + \beta\big(EV^{both} - EV^{none}\big). \end{aligned} \tag{41}$$

The first pins $\kappa^{internal}$. The second pins $\kappa^{external}$; that is the external-only share, 3.2 percent in Table 7.[25] The third, net of additivity of the two single-mode costs, pins $\kappa^{complement}$; that is the joint-mode share, 27.7 percent in the same table.

Continuation values come from the controlled law of motion for productivity. Table 4 reports mode parameters α_5 through α_8 between 0.345 and 0.390. A Wald test of equal incremental returns has $p = 0.54$. Because those parameters are close, EV^a differs little across modes, and those

[25] The Community Innovation Survey records whether the firm contracted out R&D, not the size of the contract. A small purchase and a fully outsourced agenda are the same cell.

inversions are read mainly as current costs. If those parameters differed more, the same choice frequencies would imply smaller cost gaps.

The nested fixed-point uses the conditional choice probabilities $P(a_{it} \mid s_{it})$. Because the three costs are constants and the productivity parameters are close, the same ranking appears in the unconditional shares in Table 7. Matching those shares to Table 5 gives $-\log(0.324/0.368) \approx 0.13$ next to $\kappa^{internal} = 0.127$, $-\log(0.032/0.368) \approx 2.44$ next to $\kappa^{external} = 2.453$, and $-\log(0.277/0.368) \approx 0.28$ next to the net cost of both, 0.285.

Table 2 is the reduced-form analogue. Conditional on year and sub-industry fixed effects, last period's mode raises the probability of the same mode by 42.02 percentage points for no R&D, 27.88 for internal-only, 17.80 for external-only, and 35.38 for both. External-only is the weakest own-lag, which is what a rare cell looks like in the linear probability.

The three costs are identified under the normalizations above. The bargaining share and the surplus scale in Section 3.3 are a translation of $\kappa^{external}$ and $\kappa^{complement}$ once a scale is chosen; I do not estimate them.

J Grossman–Hart algebra and observational equivalence

This appendix fills in the contracting subperiod of Section 3.3. The first block derives the equilibrium net, the efficient benchmark, and the location of $\kappa^{external}$ and $\kappa^{complement}$ in the Bellman. The second block shows why the likelihood identifies only the costs. The last subsection inverts the estimated costs at each surplus scale S. This is a reading of the two estimated costs, and the model is not re-solved.

J.1 Equilibrium net

The firm solves

$$\max_{I \geq 0} \ \tau\theta I - I^2, \qquad \tau \in (0,1],\ \theta > 0. \tag{42}$$

Differentiate with respect to I:

$$\frac{\partial}{\partial I}\left(\tau\theta I - I^2\right) = \tau\theta - 2I. \tag{43}$$

Set the derivative to zero:

$$\tau\theta - 2I = 0 \quad \Longrightarrow \quad I^* = \frac{\tau\theta}{2}. \tag{44}$$

The second derivative is

$$\frac{\partial^2}{\partial I^2}\left(\tau\theta I - I^2\right) = -2 < 0, \tag{45}$$

so the critical point is a maximum. At $I = 0$ the first derivative equals $\tau\theta > 0$, so the firm strictly prefers a positive investment. Substitute I^* into the objective:

$$\text{net}(\tau) = \tau\theta \cdot \frac{\tau\theta}{2} - \left(\frac{\tau\theta}{2}\right)^2 = \frac{\tau^2\theta^2}{2} - \frac{\tau^2\theta^2}{4} = \frac{\tau^2\theta^2}{4}. \tag{46}$$

That is equation (10) in the text.

Efficient investment. If the firm kept the entire pie, the same program would be run at $\tau = 1$. Then $I^{\mathrm{e}} = \theta/2$ and

$$S := \mathrm{net}(1) = \frac{\theta^2}{4}. \tag{47}$$

The equilibrium net is the fraction τ^2 of that surplus:

$$\mathrm{net}(\tau) = S\tau^2. \tag{48}$$

The special case $\theta = 1$ yields $\mathrm{net}(\tau) = \tau^2/4$ and $S = 1/4$. An external cost measured in millions of euro cannot be generated from $S = 1/4$, so the paper keeps θ (equivalently S) free.

An isomorphic technology. The same net is obtained from a concave surplus $\theta\sqrt{\tilde{I}}$ with linear cost $\tilde{I}$. The program is $\max_{\tilde{I}\geq 0} \tau\theta\sqrt{\tilde{I}} - \tilde{I}$. The first-order condition is $\tau\theta/(2\sqrt{\tilde{I}}) = 1$, so $\tilde{I}^* = (\tau\theta/2)^2$ and

$$\mathrm{net}(\tau) = \tau\theta \cdot \frac{\tau\theta}{2} - \Big(\frac{\tau\theta}{2}\Big)^2 = \frac{\tau^2\theta^2}{4}. \tag{49}$$

I use the quadratic-cost version in the main text because the first-order condition is linear in I. Nothing in the Bellman depends on which of the two technologies one writes down.

Location identities. Measure costs against S:

$$\kappa^{external} = \mathrm{net}(1) - \mathrm{net}(\tau) = S - S\tau^2 = S\big(1 - \tau^2\big), \tag{50}$$

$$\kappa^{complement} = \mathrm{net}(\tau') - \mathrm{net}(\tau) = S\big((\tau')^2 - \tau^2\big). \tag{51}$$

The current-period payoffs relative to the efficient net are

$$v^{external} = \mathrm{net}(\tau) - \mathrm{net}(1) = -\kappa^{external}, \tag{52}$$

$$v^{both} = -\kappa^{internal} + \mathrm{net}(\tau') - \mathrm{net}(1). \tag{53}$$

The second line expands as

$$\begin{aligned} \mathrm{net}(\tau') - \mathrm{net}(1) &= \big(\mathrm{net}(\tau') - \mathrm{net}(\tau)\big) + \big(\mathrm{net}(\tau) - \mathrm{net}(1)\big) \\ &= \kappa^{complement} - \kappa^{external}, \end{aligned} \tag{54}$$

and therefore

$$v^{both} = -\kappa^{internal} - \kappa^{external} + \kappa^{complement}, \tag{55}$$

which is the current-period term already in (5). Replacing $\mathrm{net}(\tau') - \mathrm{net}(1)$ by $\mathrm{net}(\tau') - \mathrm{net}(\tau)$ would drop $-\kappa^{external}$ and would not match the Bellman.

Underinvestment and transfer. Social surplus at investment I is $\theta I - I^2$. At the efficient investment this equals S. At $I^* = \tau\theta/2$ it equals $S(2\tau - \tau^2)$. The deadweight loss is therefore

$$\mathrm{DWL}(\tau) = S - S(2\tau - \tau^2) = S(1 - \tau)^2. \tag{56}$$

The partner collects share $1 - \tau$ of the realized pie θI^* and invests nothing, so the transfer is

$$(1 - \tau)\theta I^* = (1 - \tau)\tau \cdot 2S = 2S\tau(1 - \tau). \tag{57}$$

Adding the two pieces recovers the firm's private gap:

$$S(1-\tau)^2 + 2S\tau(1-\tau) = S(1-\tau^2) = \kappa^{external}. \tag{58}$$

The split between deadweight loss and transfer depends on S. The sum does not.

J.2 Two costs and three primitives

The identities (12)–(13) are two equations in (S, τ, τ'). They do not pin down a unique bargaining share. For any surplus scale $S > \kappa^{external}$,

$$\tau = \sqrt{1 - \kappa^{external}/S}, \qquad \tau' = \sqrt{1 - (\kappa^{external} - \kappa^{complement})/S}. \tag{59}$$

The ratio of the two costs does not depend on S:

$$r := \frac{\kappa^{complement}}{\kappa^{external}} = \frac{(\tau')^2 - \tau^2}{1 - \tau^2}. \tag{60}$$

On the set consistent with a given r, $\tau' \geq \sqrt{r}$. The likelihood therefore recovers the three costs, not τ. Table 6 in Section 5 reports the sample ratio, and Appendix J.4 inverts the two costs over a range of S.

J.3 Why the likelihood identifies only the costs

Write $\overline{V}^a$ for the deterministic alternative-specific value of mode a in (5), omitting the Type-I shock. Under the estimated Bellman,

$$\begin{aligned}
\overline{V}^{\text{no}} &= \beta EV^{\text{no}}, \\
\overline{V}^{\text{internal}} &= -\kappa^{internal} + \beta EV^{\text{internal}}, \\
\overline{V}^{\text{external}} &= -\kappa^{external} + \beta EV^{\text{external}}, \\
\overline{V}^{\text{both}} &= -\kappa^{internal} - \kappa^{external} + \kappa^{complement} + \beta EV^{\text{both}}.
\end{aligned}$$

Under the hold-up model, the current-period nets for external-only and both are $v^{external}$ and v^{both} from (14)–(15). These nets are constants that depend only on the mode. The identities (12)–(13) then make those constants coincide with $-\kappa^{external}$ and $-\kappa^{internal} - \kappa^{external} + \kappa^{complement}$. Since $P(\omega' \mid s, a)$ depends on the mode dummy, and neither on I^* nor on τ, the four continuation values EV^a coincide with those of the estimated Bellman. Since disagreement payoffs after I is sunk are $(0, 0)$, continuation values do not enter the bargaining problem, so the net does not depend on V. Since the Type-I shocks are attached to the four modes, and bargaining takes place inside the chosen mode, the McFadden inclusive value is the log-sum-exp of the same four $\overline{V}^a$. Therefore the choice probabilities (8) coincide, and so does the likelihood.

The two identities are two equations in three primitives, so a surplus scale remains free. The likelihood therefore recovers the three costs, not τ.

The claim is a sufficiency statement. If productivity depended on $I^*(\tau)$, the nested continuation would move with τ and would no longer match the estimated Bellman. I also maintain that disagreement payoffs after investment is sunk are $(0, 0)$, so that the net does not depend on V.

J.4 Bargaining shares for an assumed surplus

Saying anything about τ, the firm's share when it contracts out alone, requires a value of S. Figure 3 traces Equation (59) for each period of Table 5. Each curve starts at that period's cost of external R&D, since a surplus smaller than the estimated external cost cannot generate it. To put the three curves on the same figure, S therefore has to be larger than 2.695 million EUR. Take, for example, $S = 5$ million EUR. The implied τ is then 0.74 before the expansion, 0.71 in the full sample, and 0.68 after. Of the cost of external R&D, the share that comes from underinvestment is then 0.15, 0.17, and 0.19, respectively, and the remainder is the value transferred to the partner. As for the cost of internal R&D, which falls to about zero after the expansion, the formulas for τ and τ' do not depend on it.

These numbers hold S fixed across periods. If S was indeed the same, the higher cost of external R&D after 2012 would mean a smaller share for a firm that contracts out without in-house R&D. However, the sample does not tell whether S itself changed with the Innovation Box, so I do not interpret this change as a change in bargaining power due to the policy. The ratio r in Table 6 does not require this assumption, and it is stable across periods.

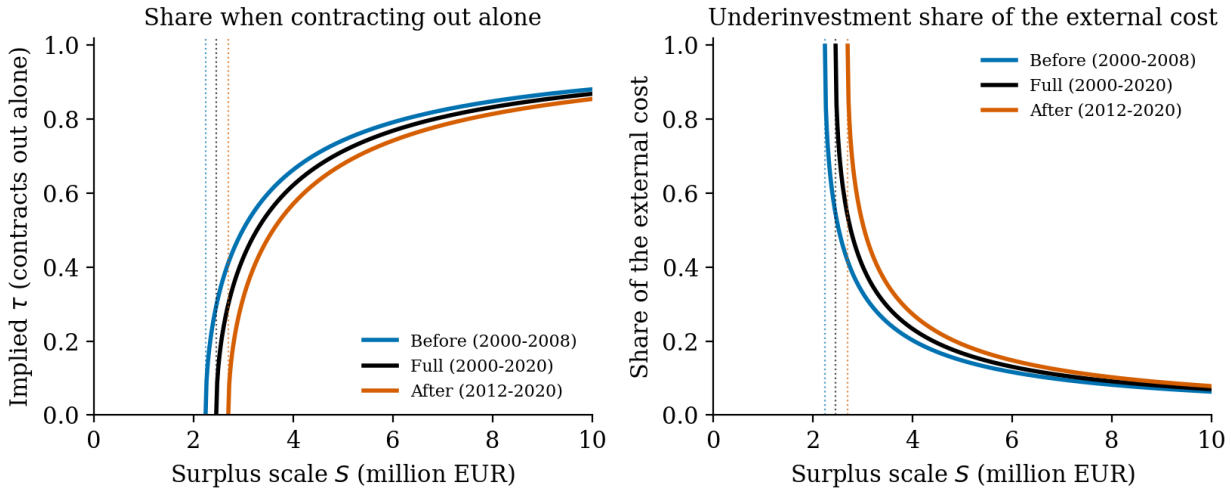


Figure 3: Left: bargaining share when the firm contracts out without in-house R&D. Right: share of the estimated external cost that is underinvestment. Each curve uses the $(\kappa^{external}, \kappa^{complement})$ pair from Table 5 for that window. Vertical dotted lines mark the three $\kappa^{external}$ thresholds. The inversion is undefined for $S \leq \kappa^{external}$.